PROJECT: BEE-1012

sub-system: EXP

Version: 1.0.0

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Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER
Design by: SANG HUYNH	Sheet title: 01_COVER PAGE.SchDoc — Sheet 1 of 33
Checked by: BAO BUI Q.	Document No: 1 — Size: A3
Approved by: VU PHAM	Revision: V1.2.2 — Date: 12/2/2025

Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER
Design by: SANG HUYNH	Sheet title: 02_BLOCK DIAGRAM.SchDoc Sheet 2 of 33
Checked by: BAO BUI Q.	Document No: 2 / Size: A3
Approved by: VU PHAM	Revision: V1.2.2 / Date: 12/6/2025

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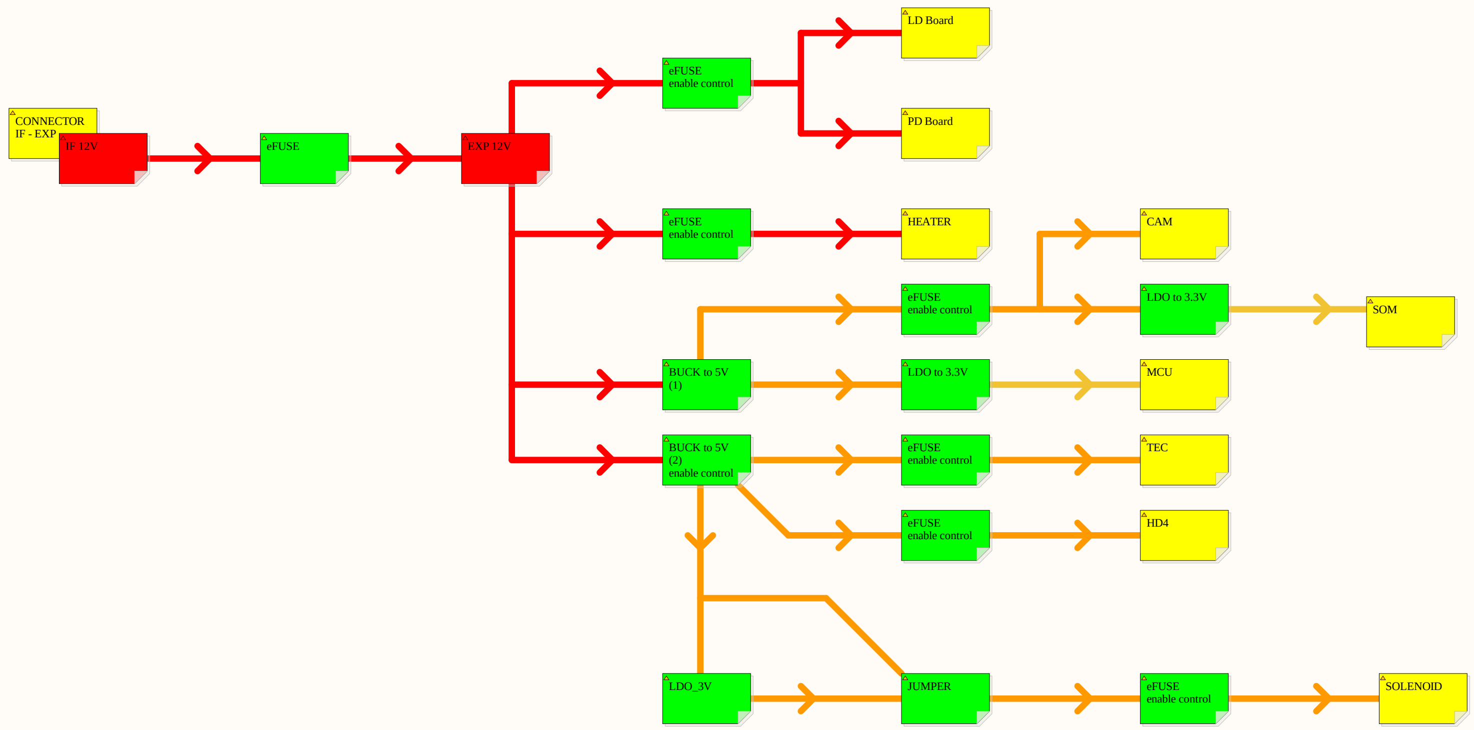
Design by: SANG HUYNH	Sheet title: 03_INDEX.SchDoc	Sheet 3 of 33
Checked by: BAO BUI Q.	Document No: 3	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 12/2/2025



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Design by: SANG HUYNH	Sheet title: 04_POWER_TREE.SchDoc	Sheet 4 of 33
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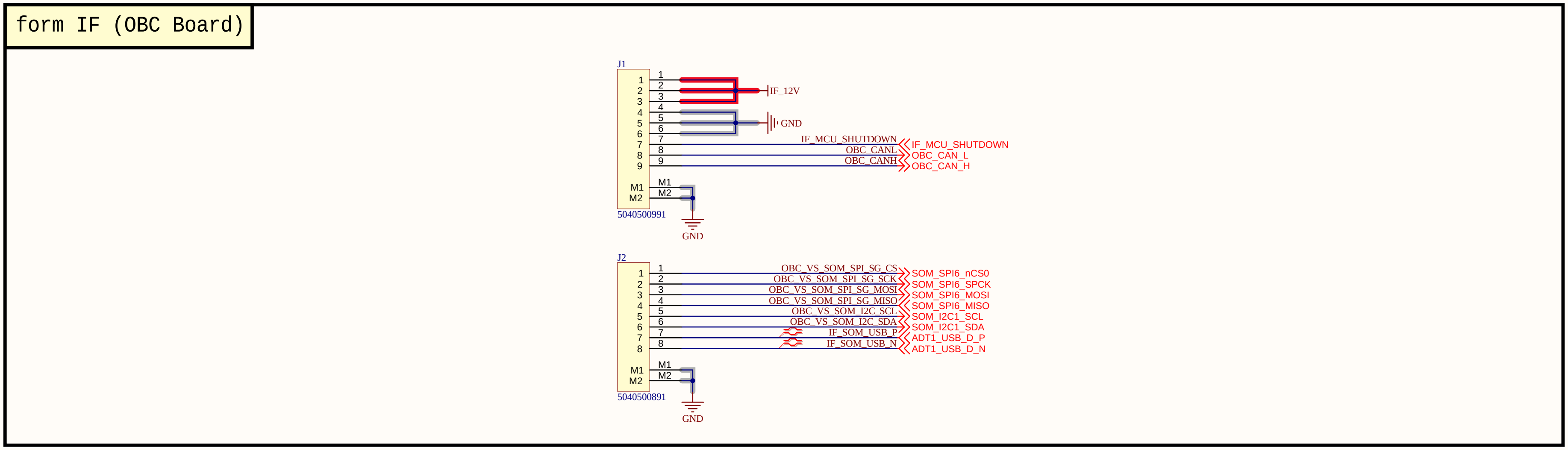
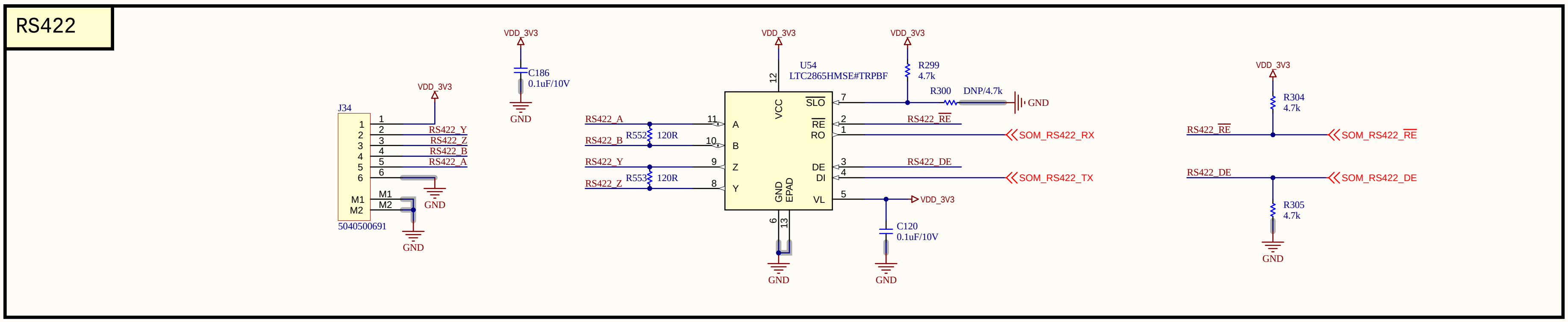
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/6/2026





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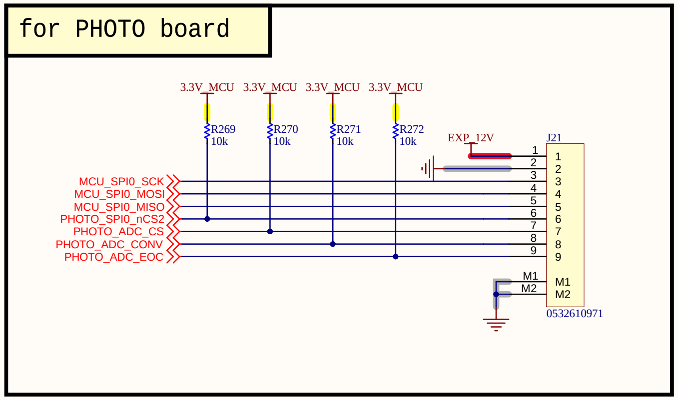
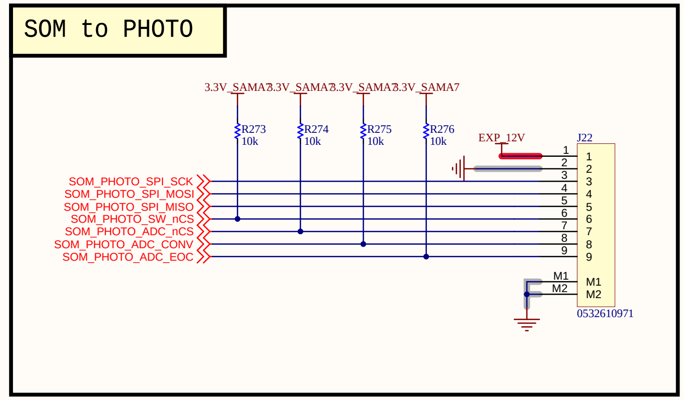
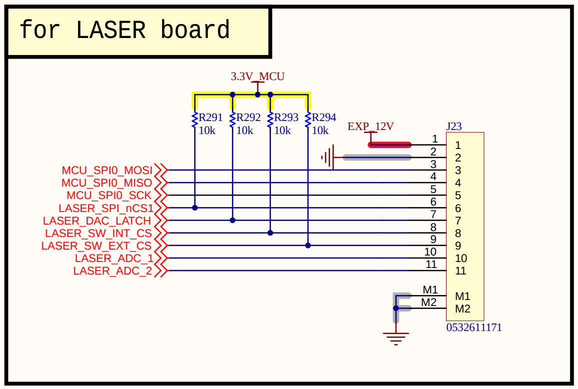
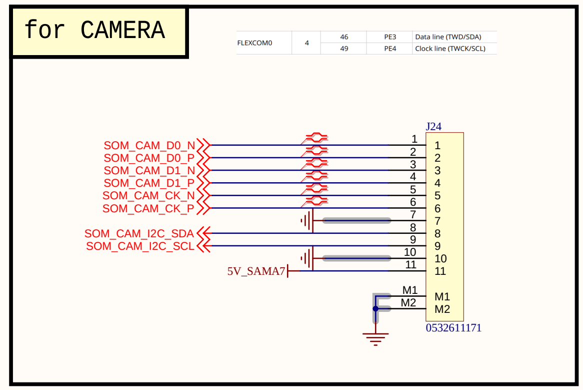
Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER		
Design by: SANG HUYNH	Sheet title: 05_CONNECTOR_IN.SchDoc	Sheet 5 of 33	
Checked by: BAO BUI Q.	Document No: 5	Size: A3	
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/12/2026	









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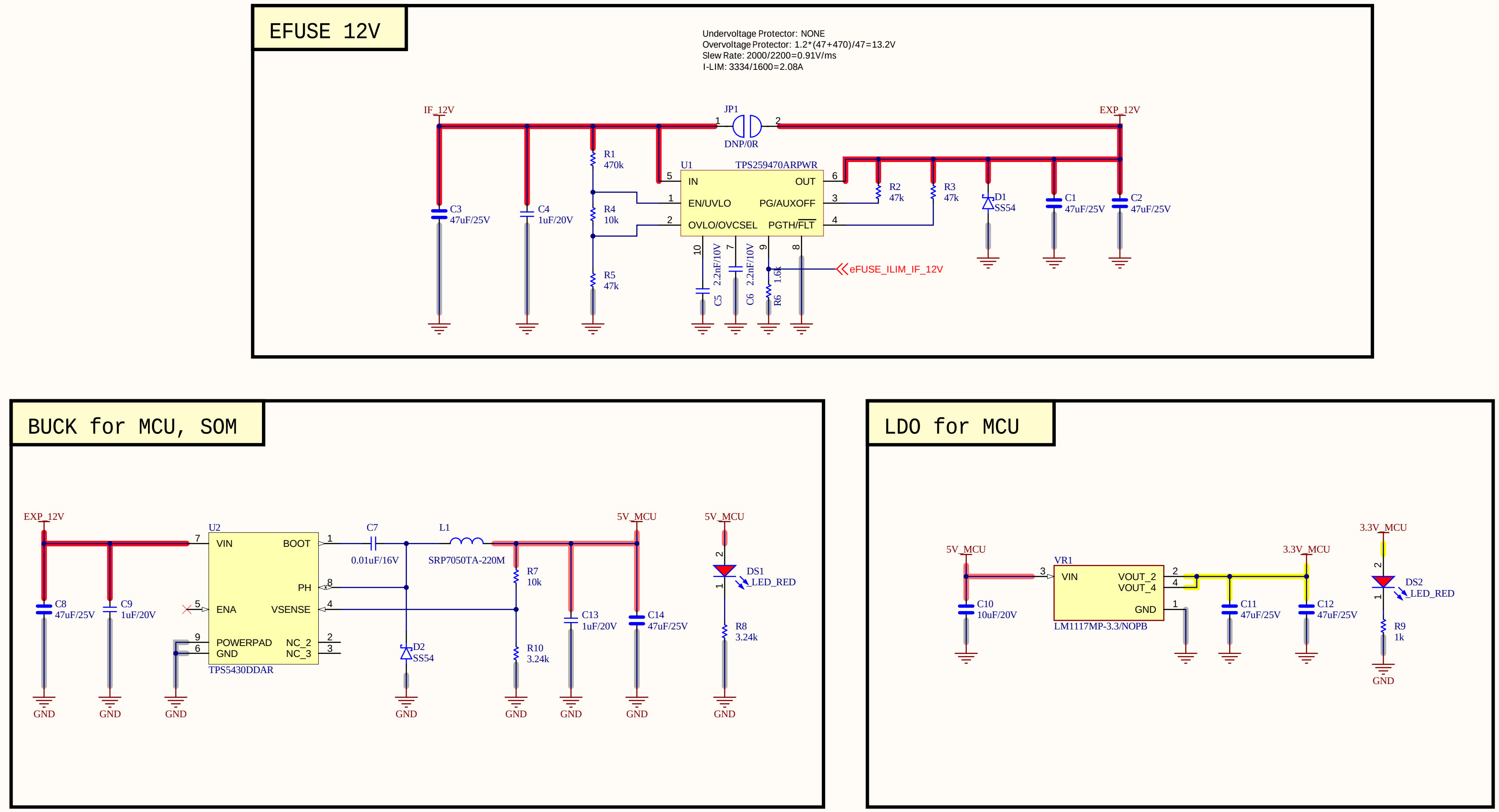

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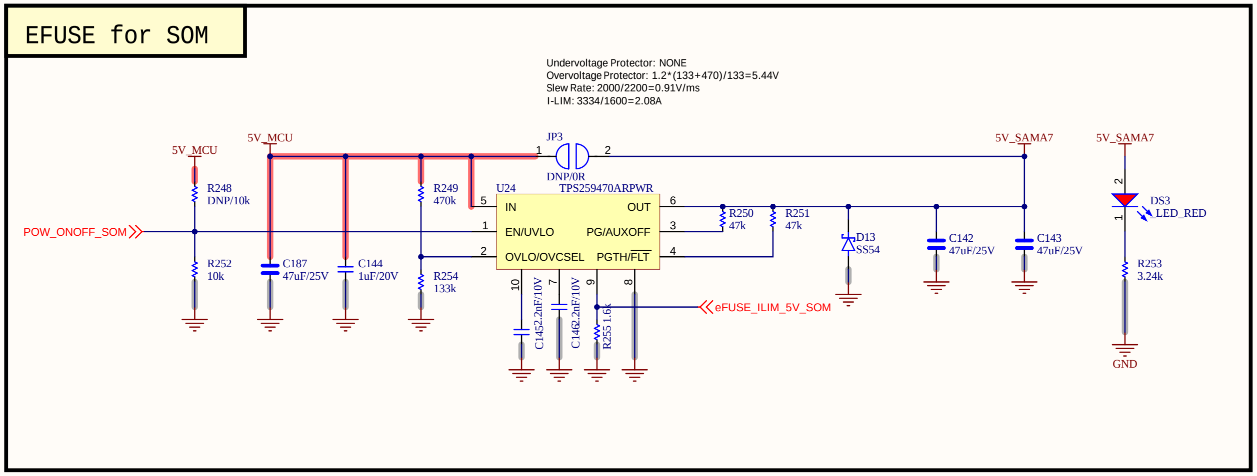
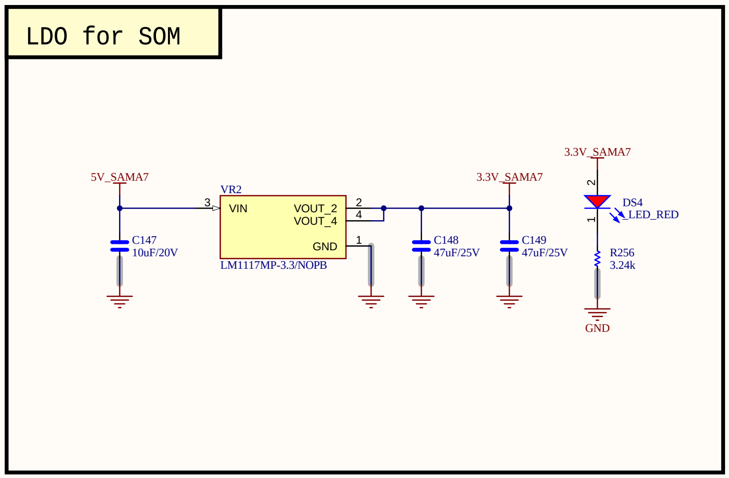
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/7/2026





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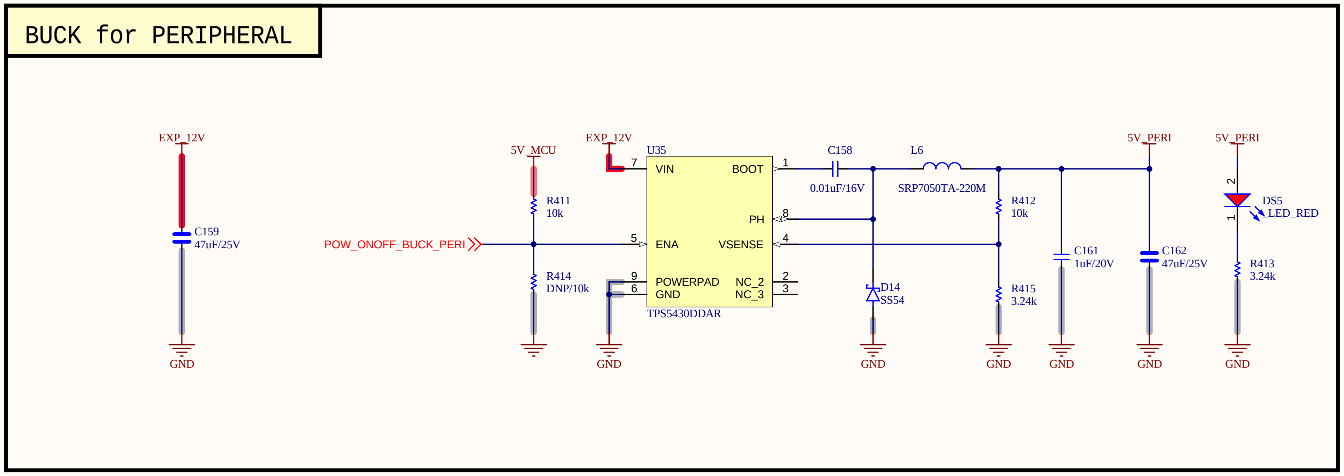
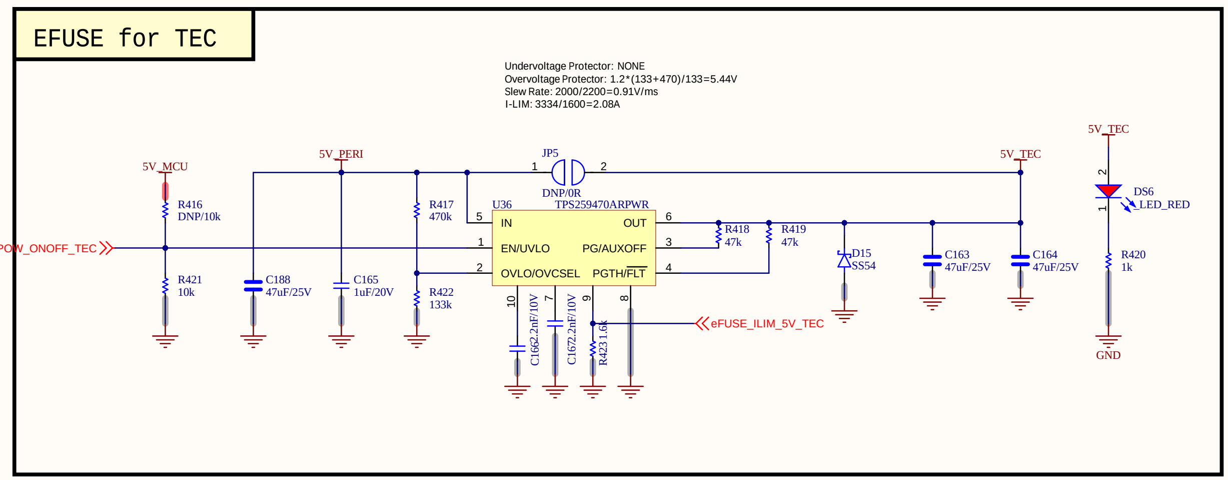
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/7/2026





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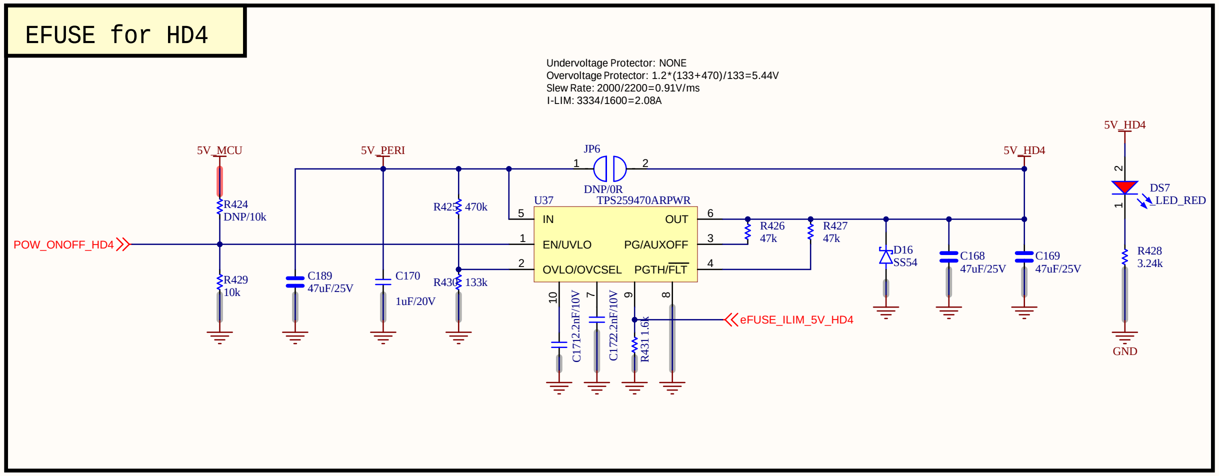

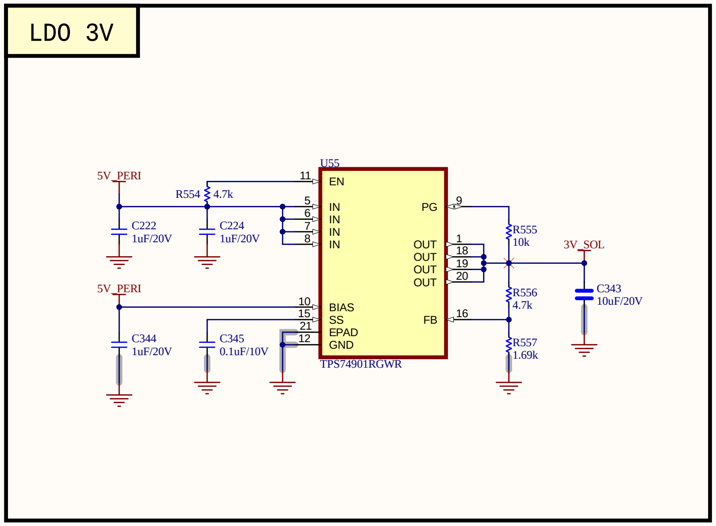

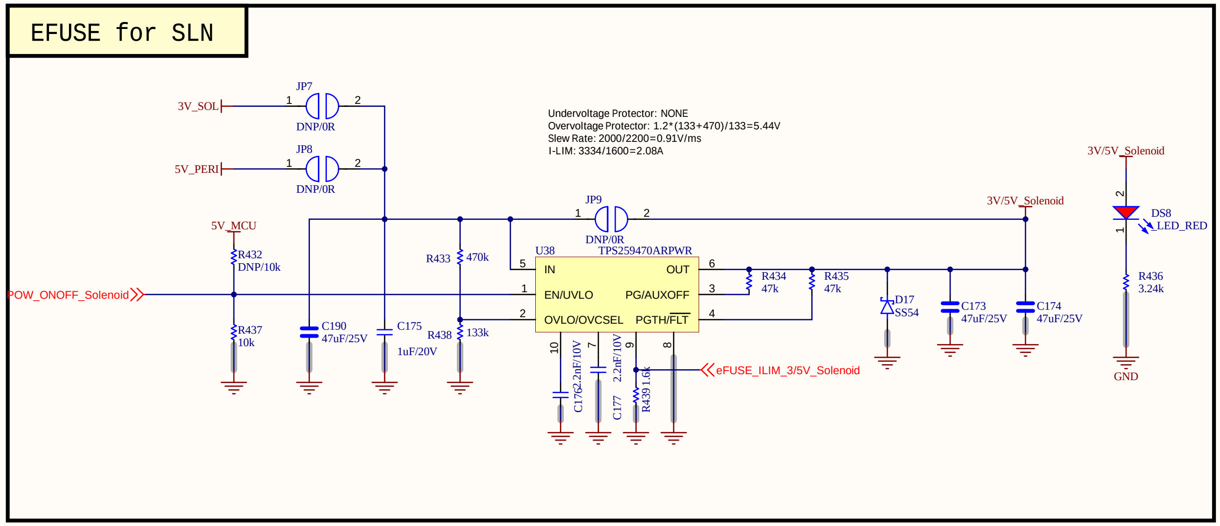

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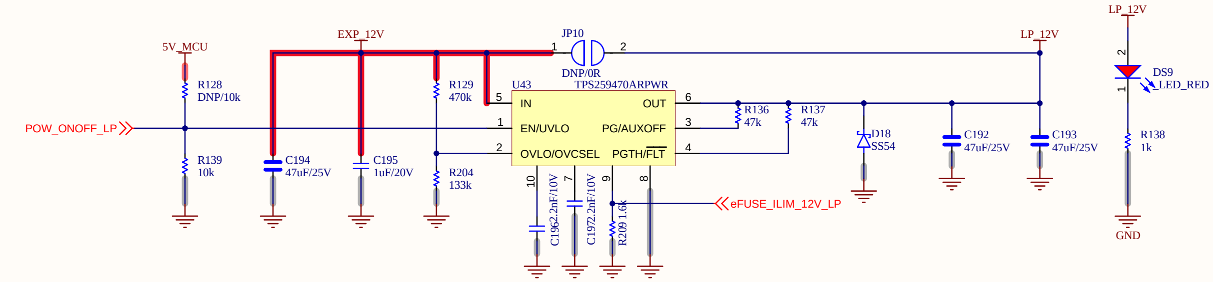
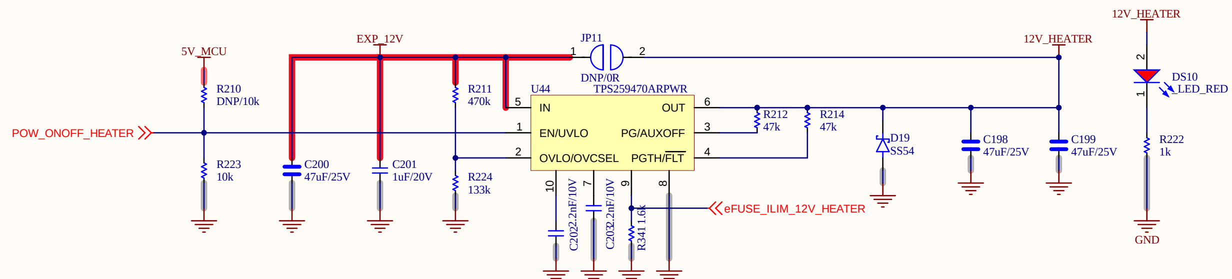
EFUSE for LP

Undervoltage Protector: NONE
Overvoltage Protector: 1.2*(133+470)/133=5.44V
Slew Rate: 2000/2200=0.91V/ms
I-LIM: 3334/1600=2.08A



EFUSE for HEATER

Undervoltage Protector: NONE
Overvoltage Protector: 1.2*(133+470)/133=5.44V
Slew Rate: 2000/2200=0.91V/ms
I-LIM: 3334/1600=2.08A



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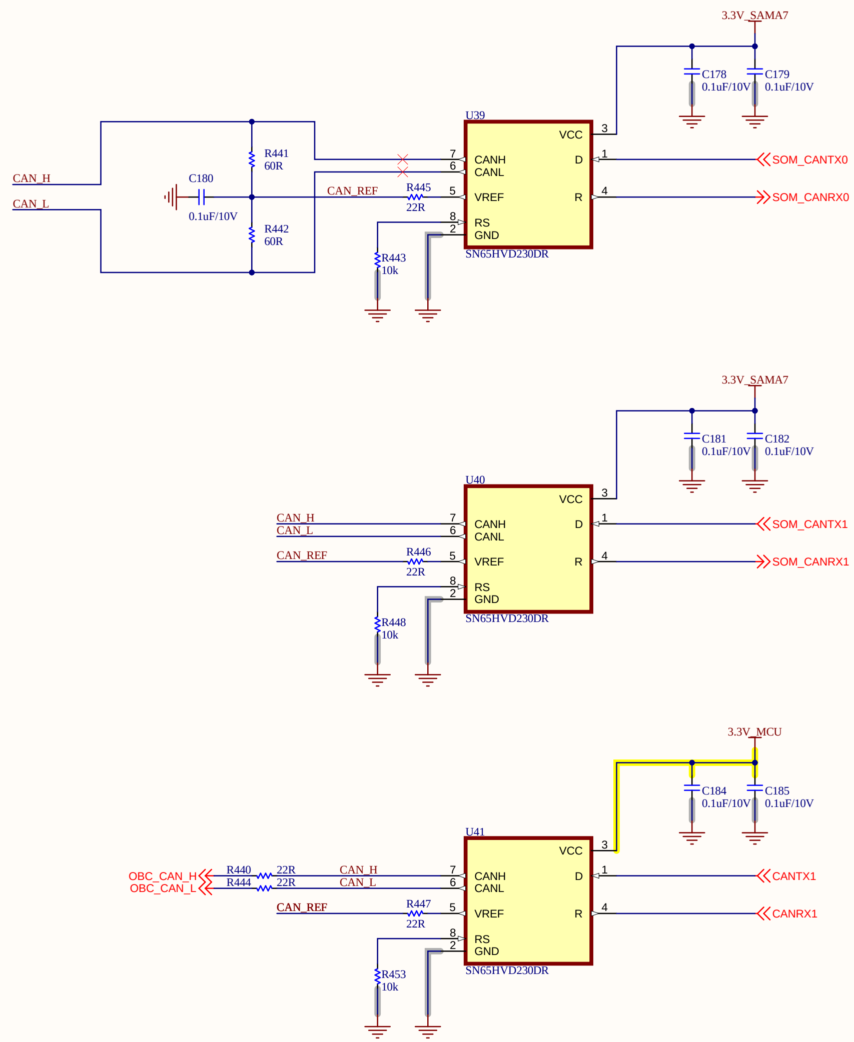
Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 06_POWER_5.SchDoc	Sheet 11 of 33
Checked by: BAO BUI Q.	Document No: 11	Size: A3
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/7/2026



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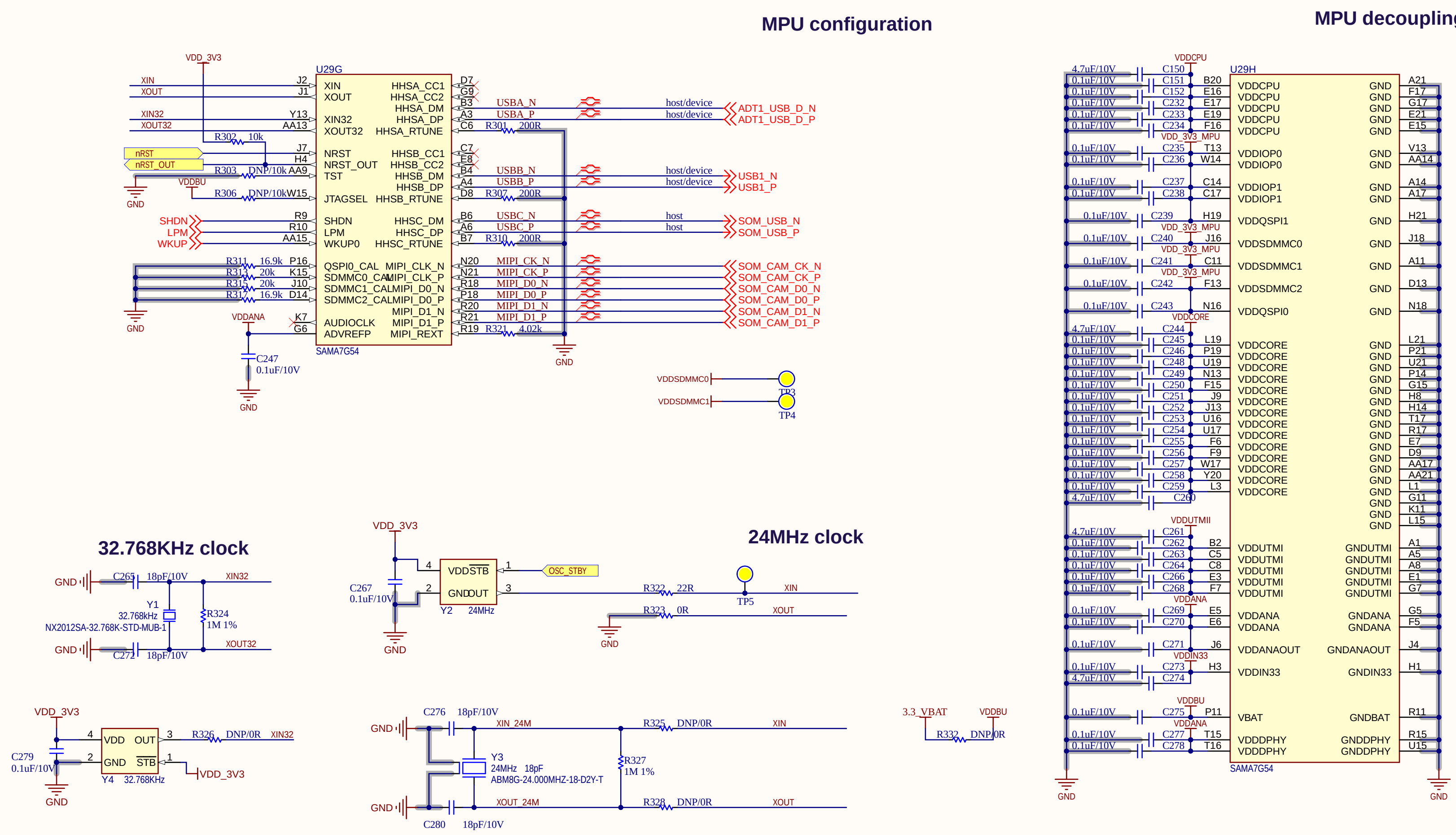
Design by: SANG HUYNH	Sheet title: 07_CAN.SchDoc	Sheet 12 of 33	
Checked by: BAO BUI Q.	Document No: 12	Size: A3	
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/9/2026	



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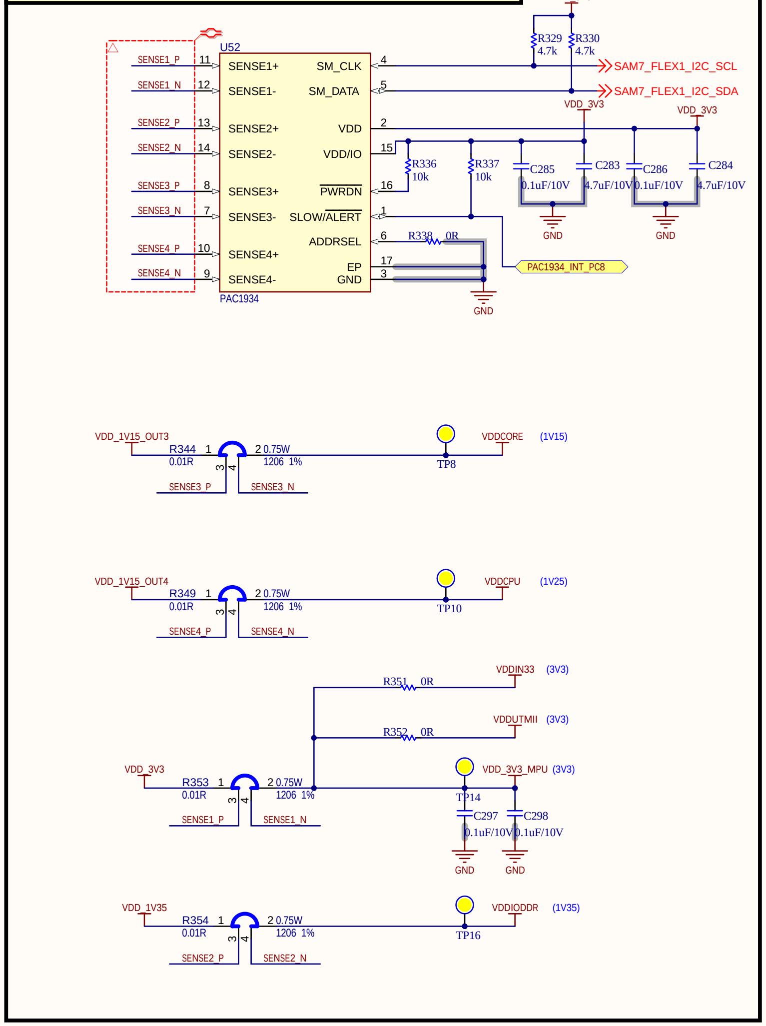
Project: BEE-PC	Sub-system: EXP	
Design by: Cam Doan	Sheet title: 08_SAMA7G54-MPU.SchDoc	Sheet 13 of 33
Checked by: Bao Bui	Document No: 13	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025

Power Measurement and Distribution



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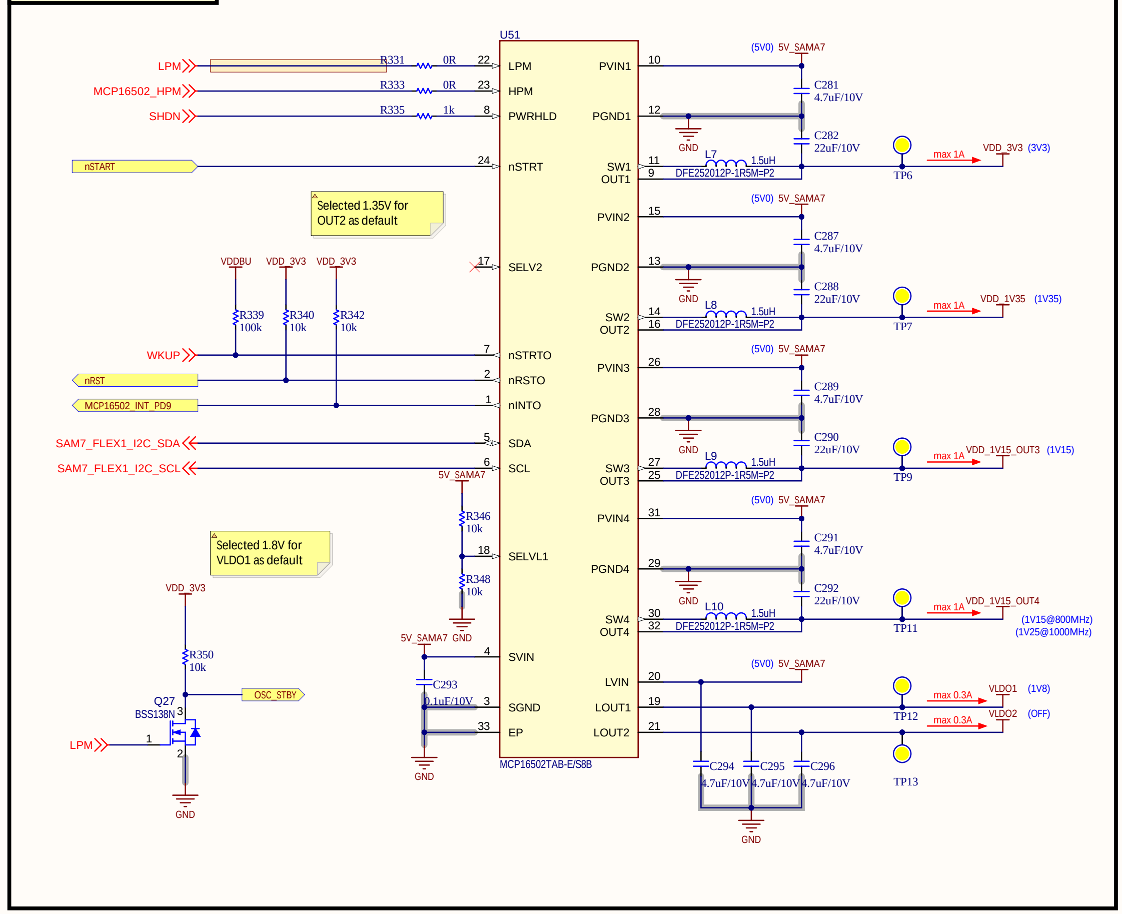

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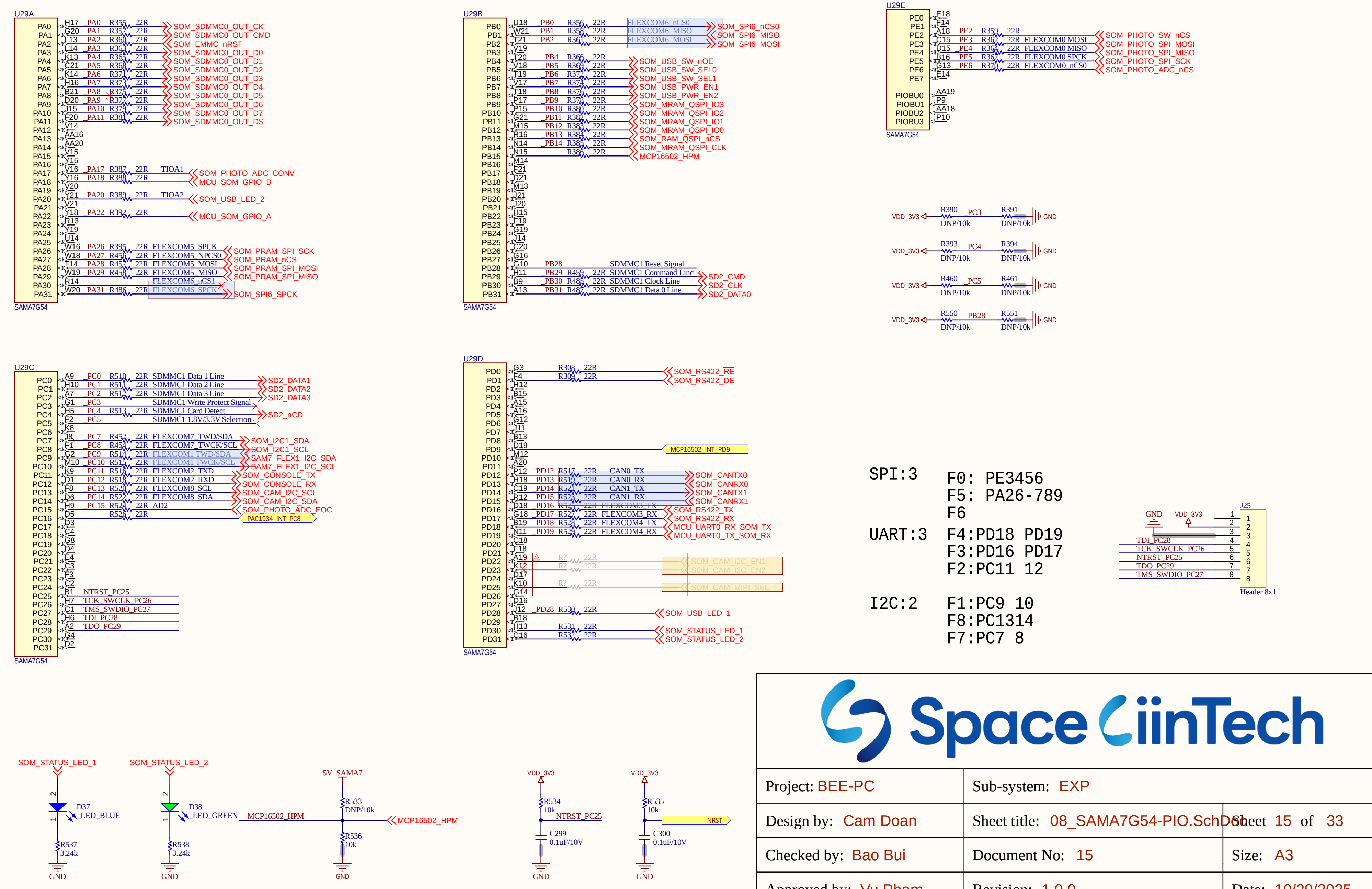
SAMA7G5 Bus Distribution



SPI:3
F0: PE3456
F5: PA26-789
F6

UART:3
F4:PD18 PD19
F3:PD16 PD17
F2:PC11 12

I2C:2
F1:PC9 10
F8:PC1314
F7:PC7 8

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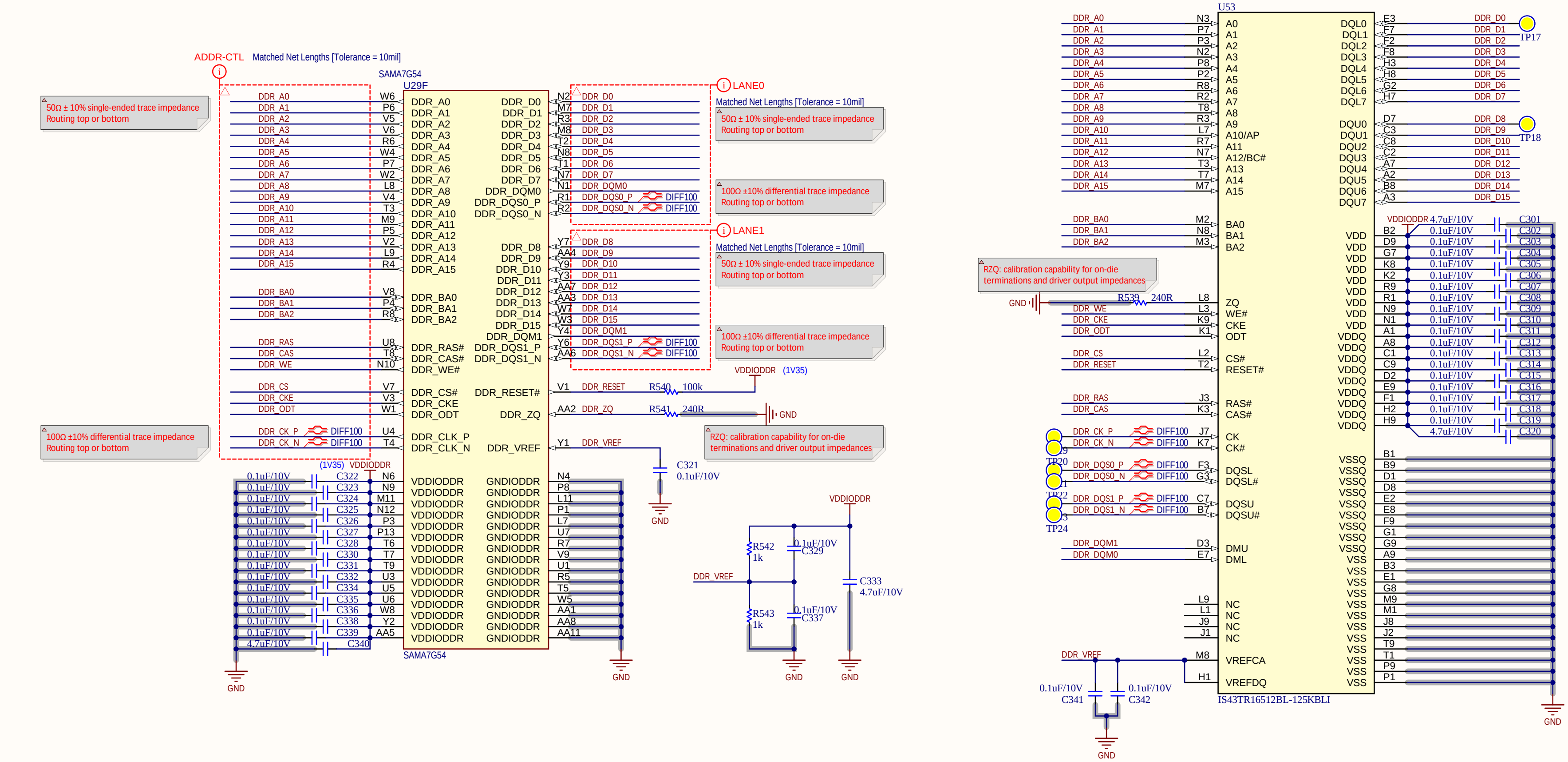
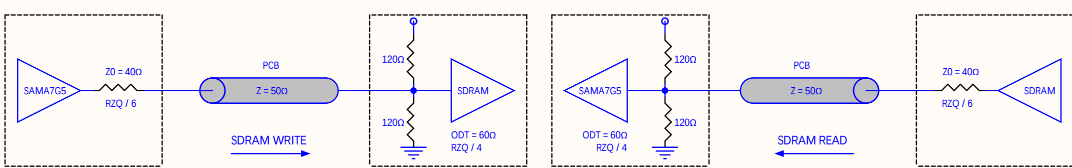
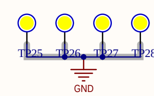
Approved by: Vu Pham	Revision: 1.0.0 — Date: 10/29/2025

SAMA7G5 DDR3-1066 controller







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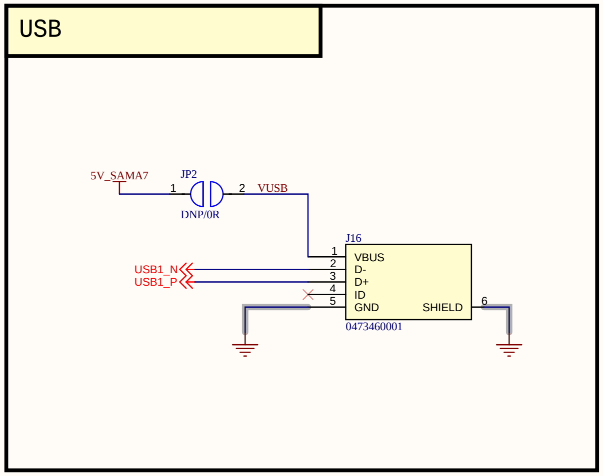
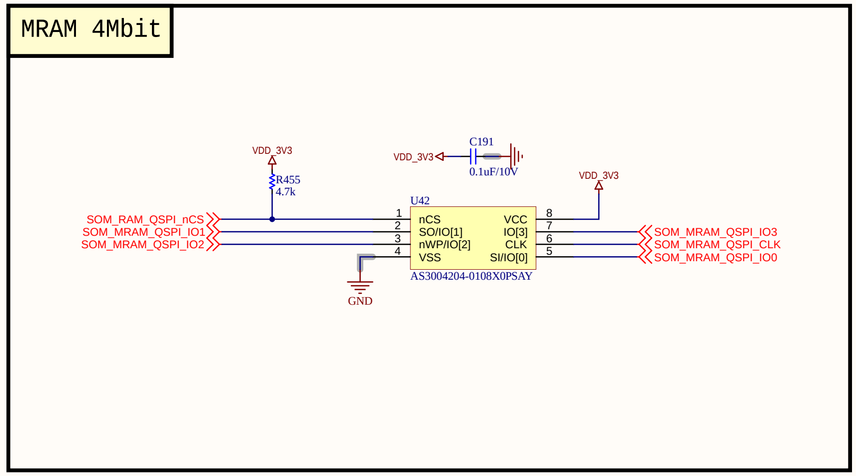
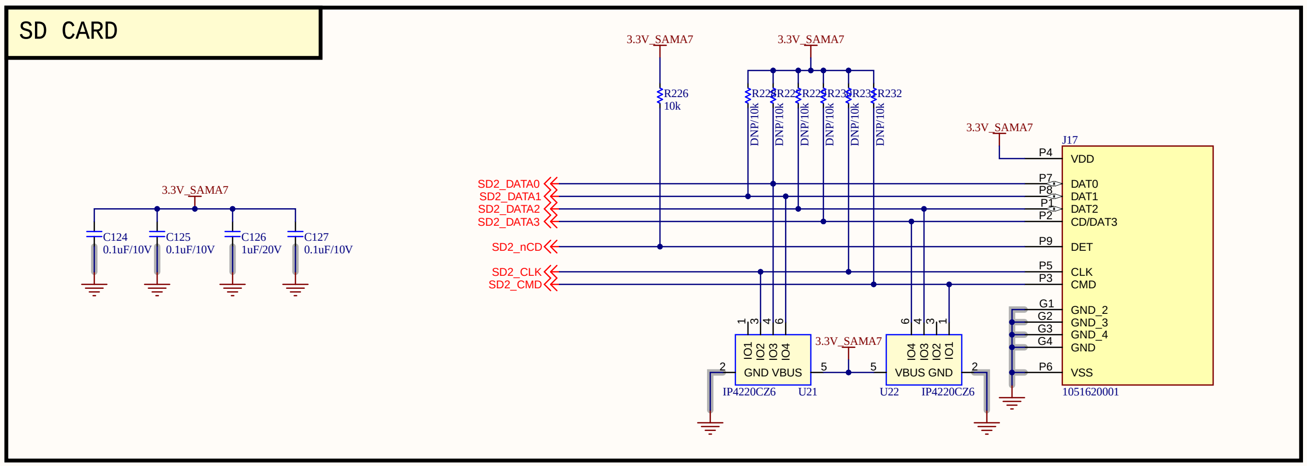
Design by: Cam Doan	Sheet title: 08_SAMA7G54-DDR3.SchDoc	Sheet 16 of 33
Checked by: Bao Bui	Document No: 16	Size: A3
Approved by: Vu Pham	Revision: 1.0.0	Date: 10/29/2025







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Design by: SANG HUYNH	Sheet title: 08_SOM_USB_CARD.SchDoc Sheet 17 of 33
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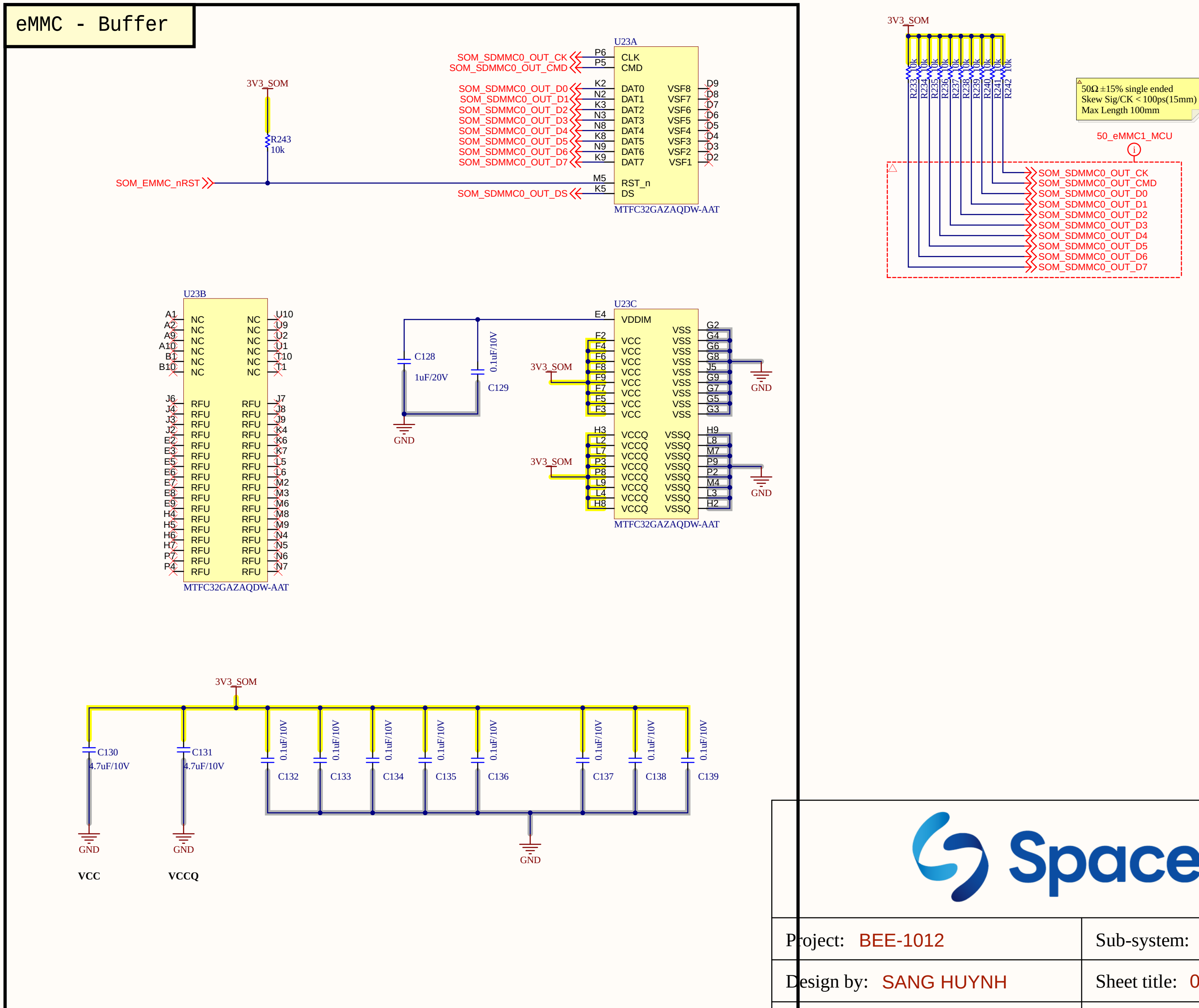
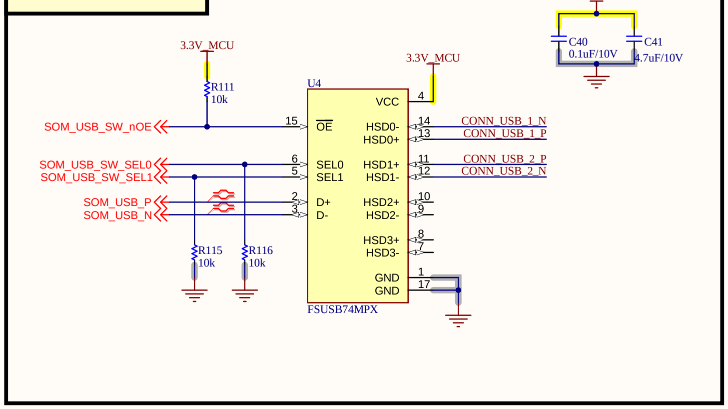
Approved by: VU PHAM	Revision: V1.2.2 Date: 1/12/2026



Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER
Design by: SANG HUYNH	Sheet title: 09_eMMC.SchDoc — Sheet 18 of 33
Checked by: BAO BUI Q.	Document No: 18 — Size: A3
Approved by: VU PHAM	Revision: V1.2.2 — Date: 1/7/2026

USB SW 1:4



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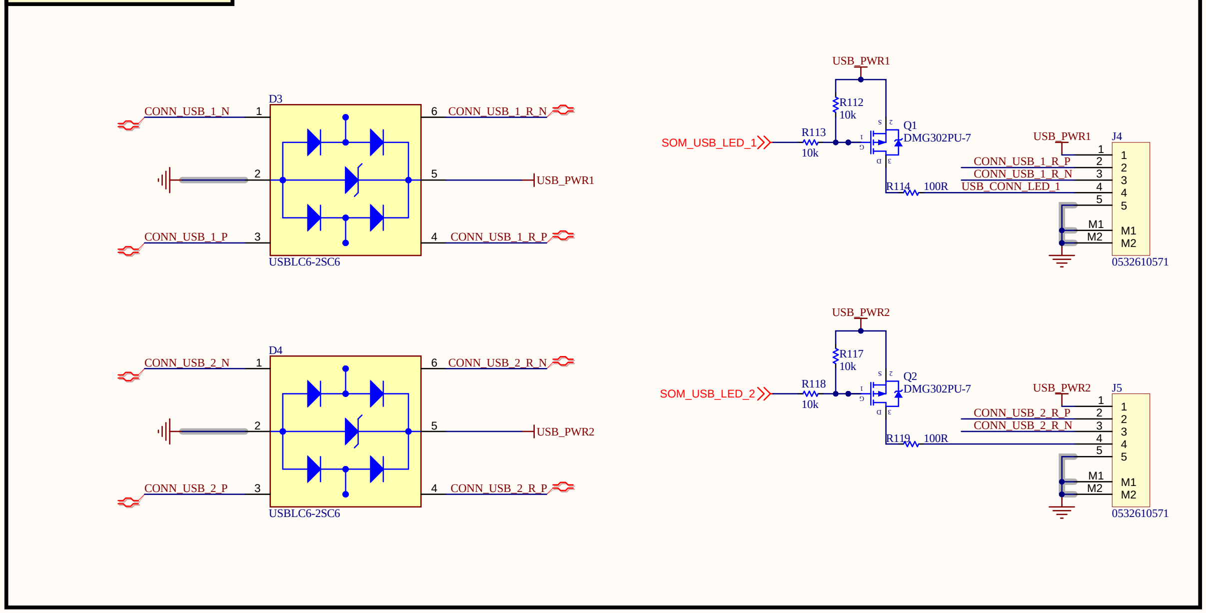

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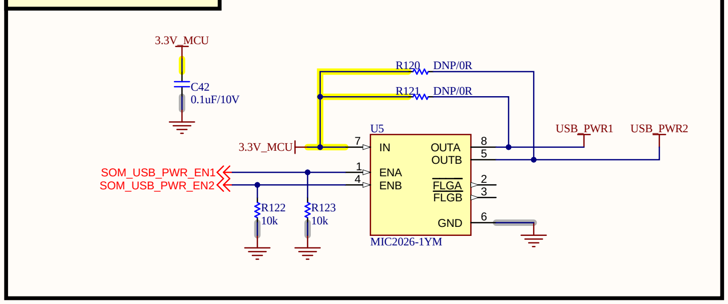

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Approved by: VU PHAM	Revision: V1.2.2	Date: 1/7/2026

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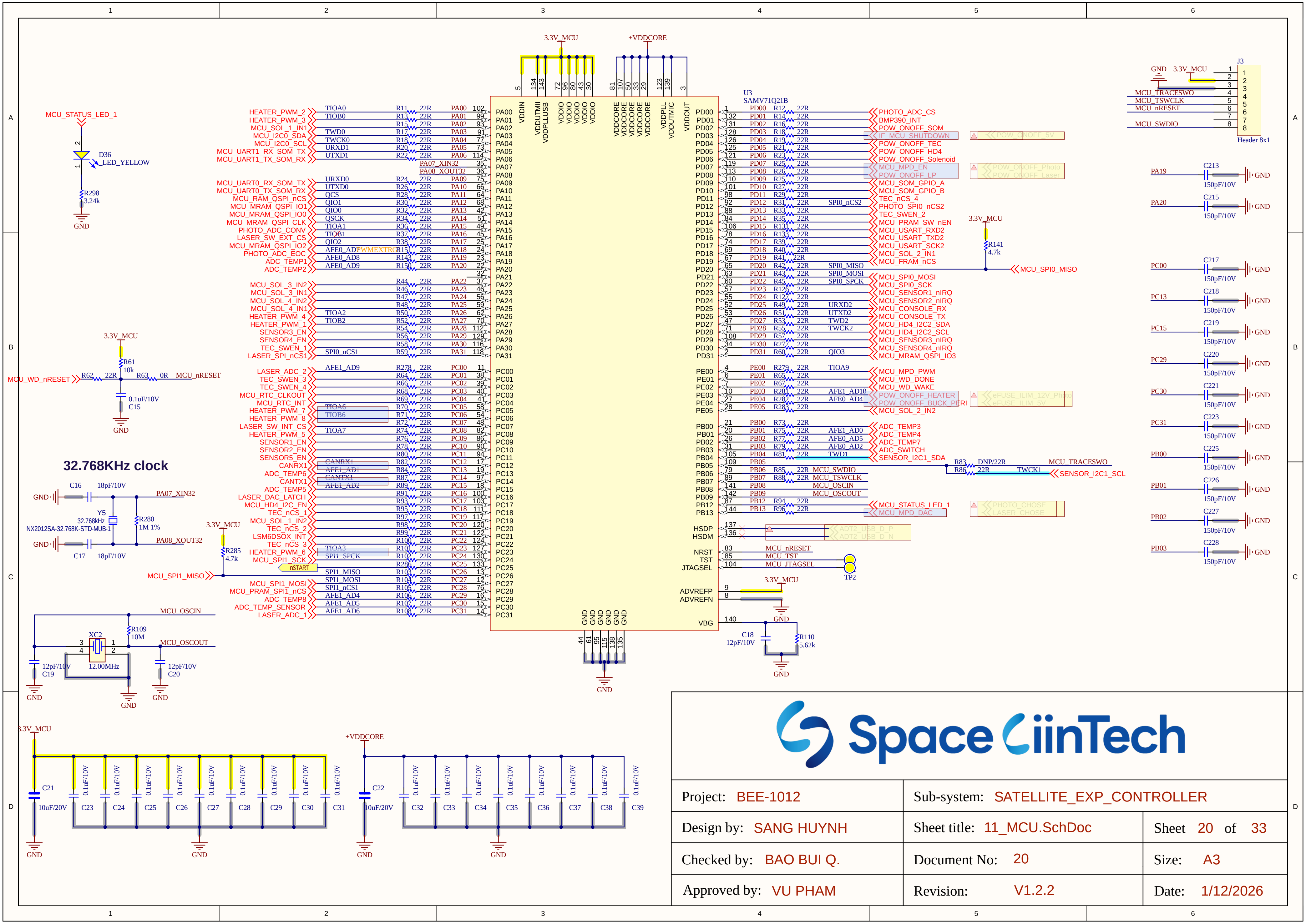

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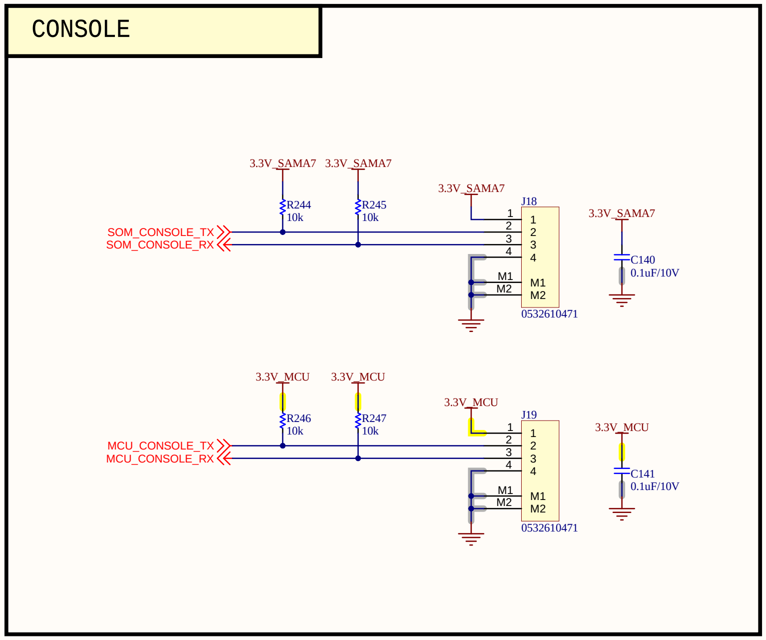

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Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
Design by: SANG HUYNH	Sheet title: 12_CONSOLE.SchDoc	Sheet 21 of 33
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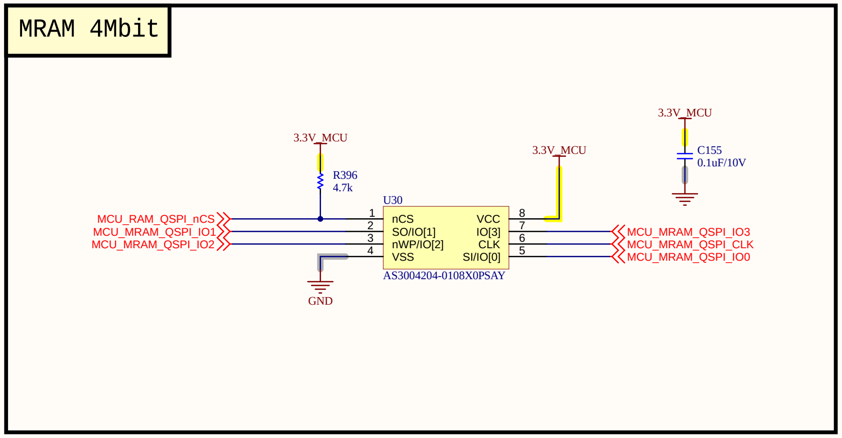
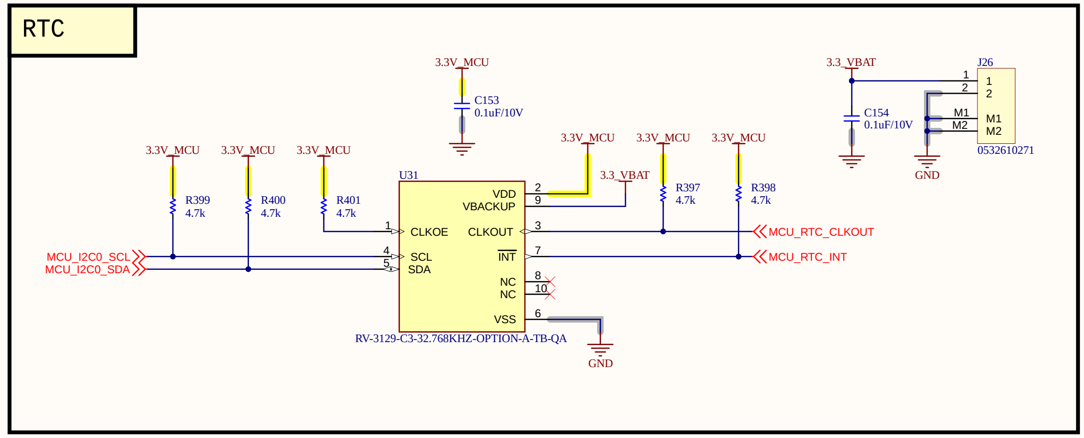
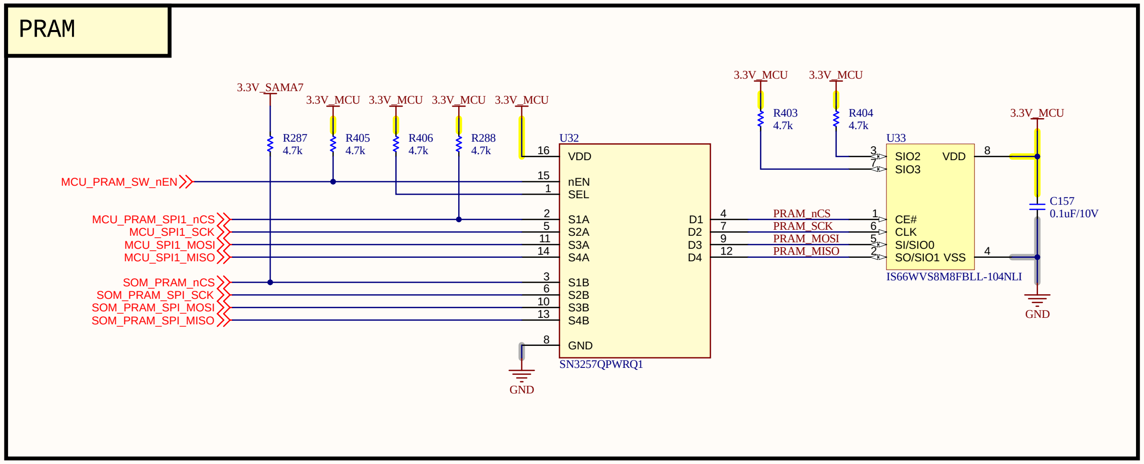
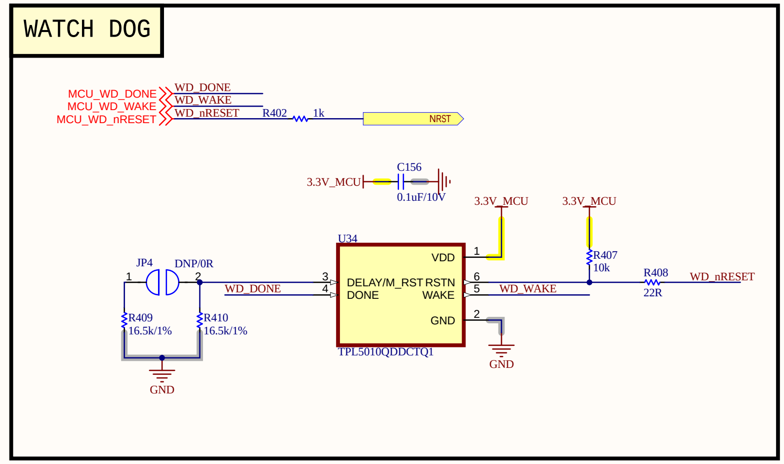
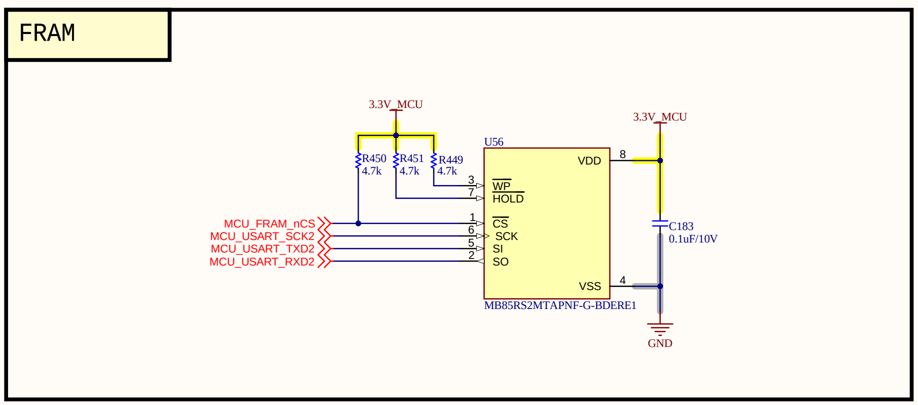
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/7/2026











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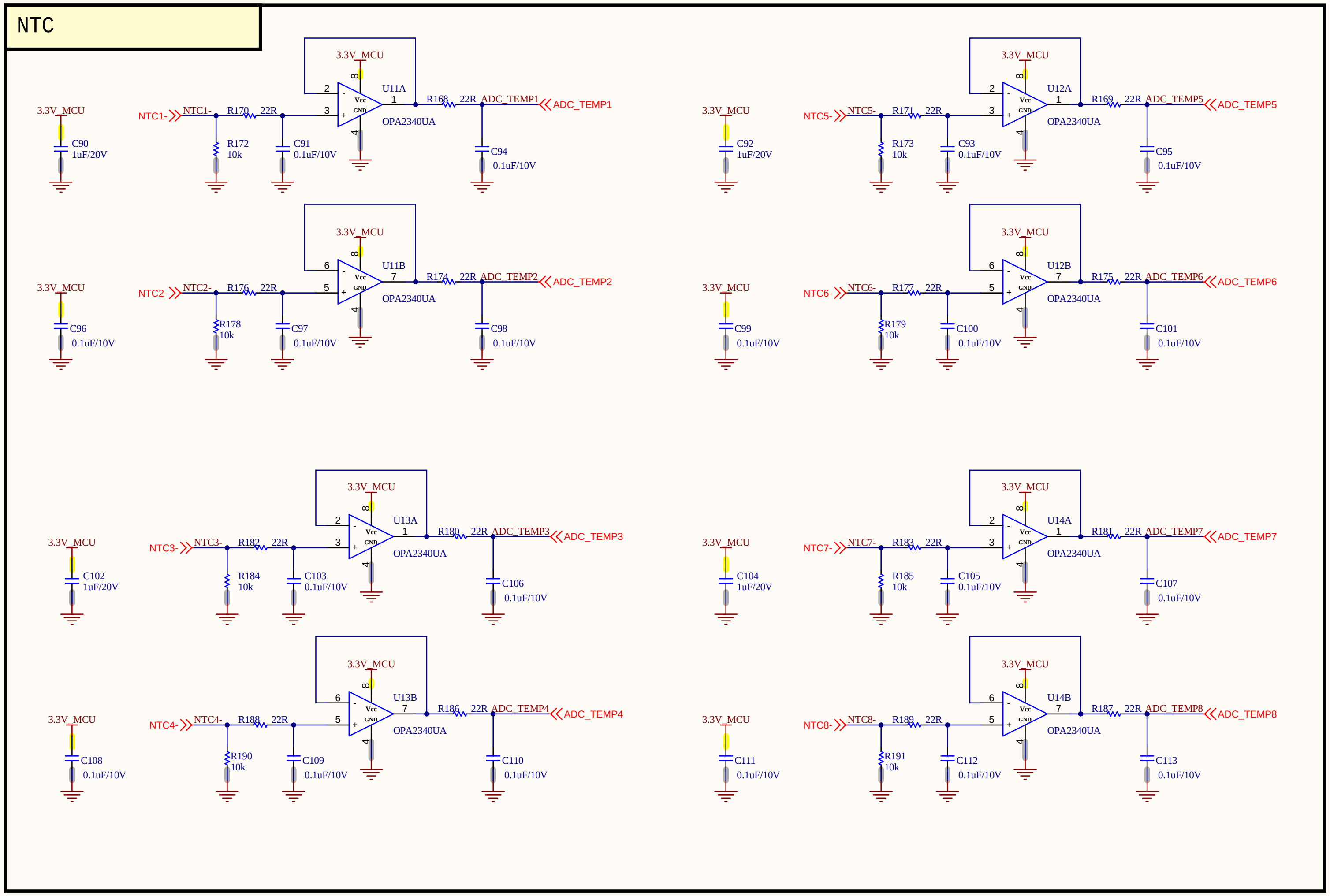
NTC



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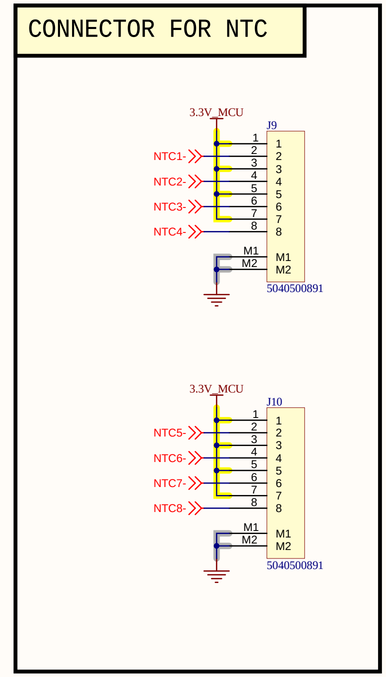

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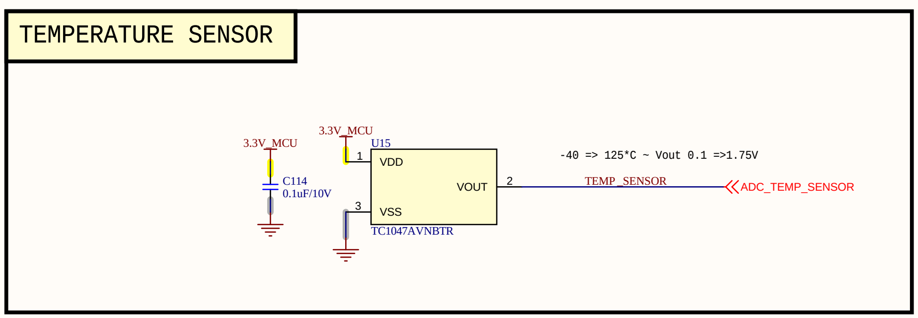

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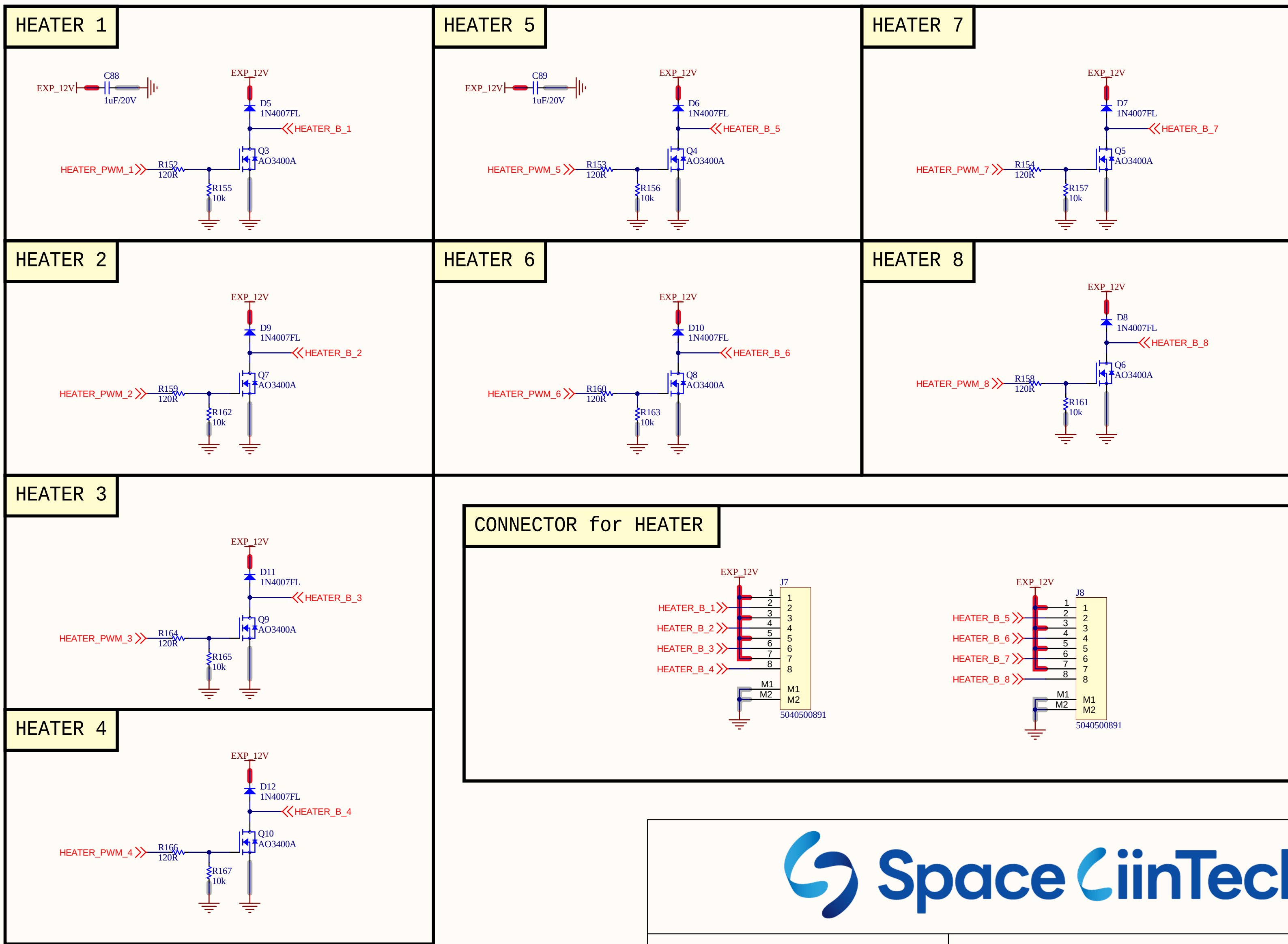

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Design by: SANG HUYNH	Sheet title: 15_HEATER.SchDoc	Sheet 24 of 33
Checked by: BAO BUI Q.	Document No: 24	Size: A3
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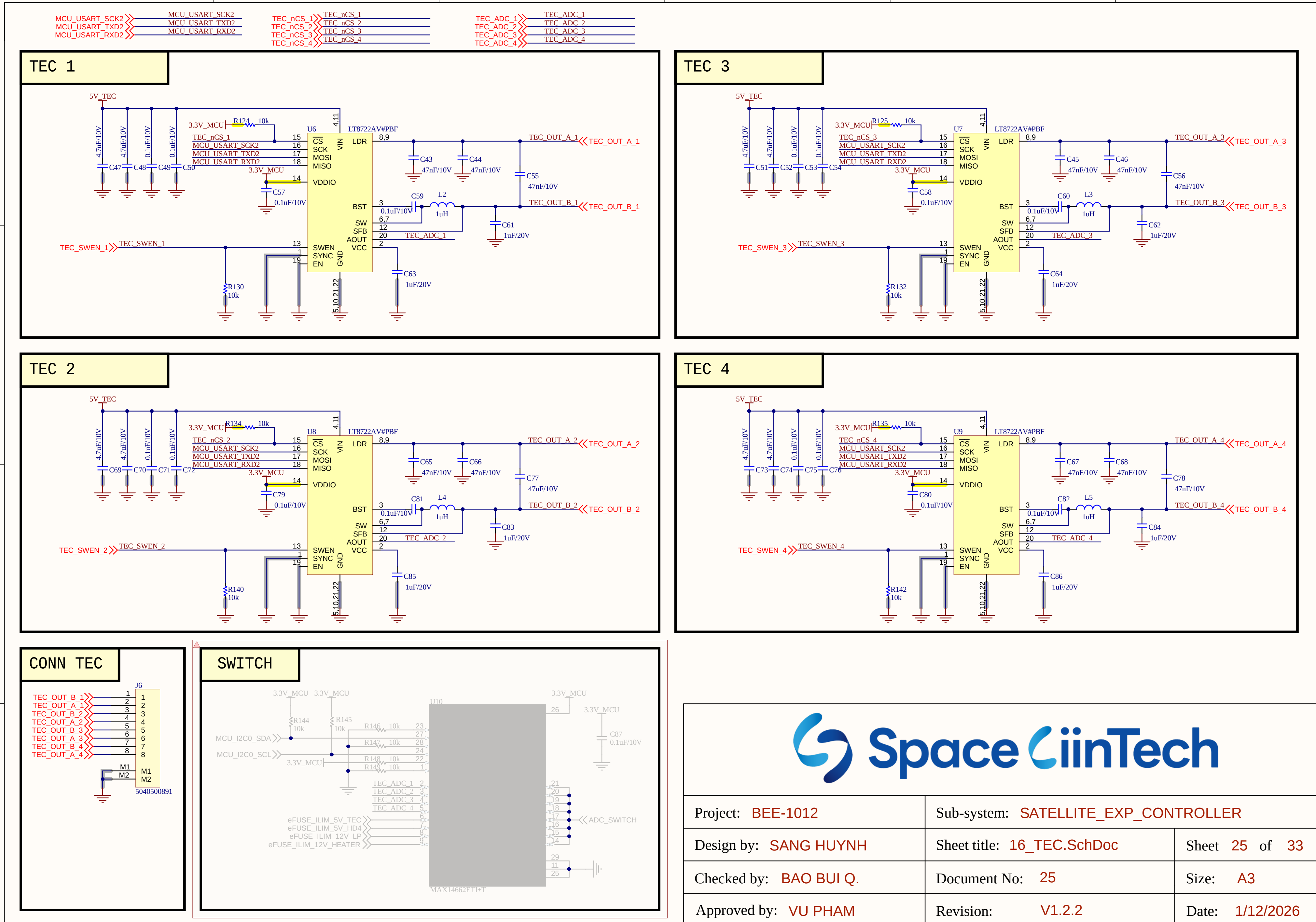

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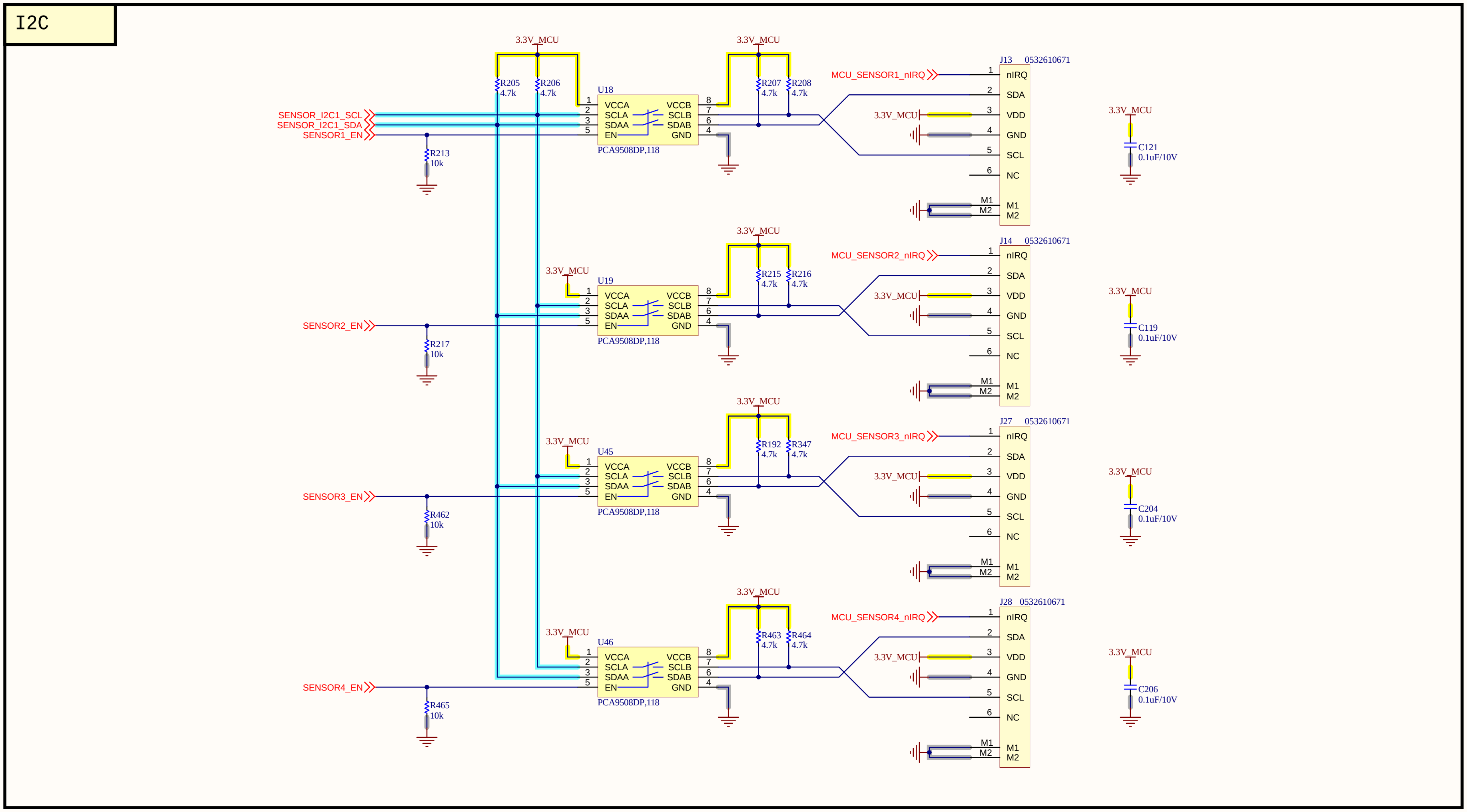
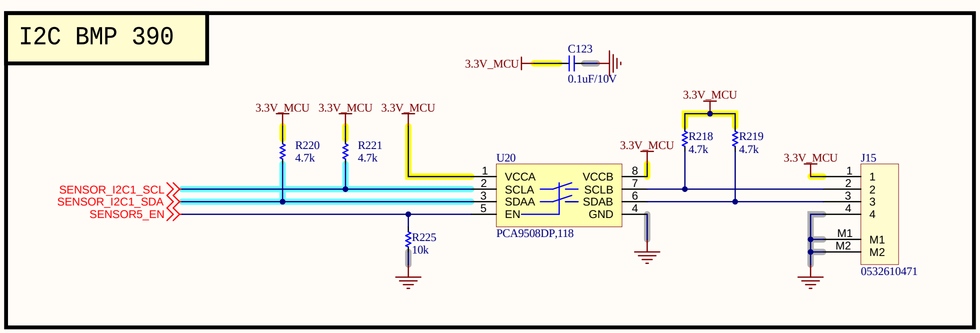
Approved by: VU PHAM	Revision: V1.2.2	Date: 1/12/2026





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Project: BEE-1012	Sub-system: SATELLITE_EXP_CONTROLLER	
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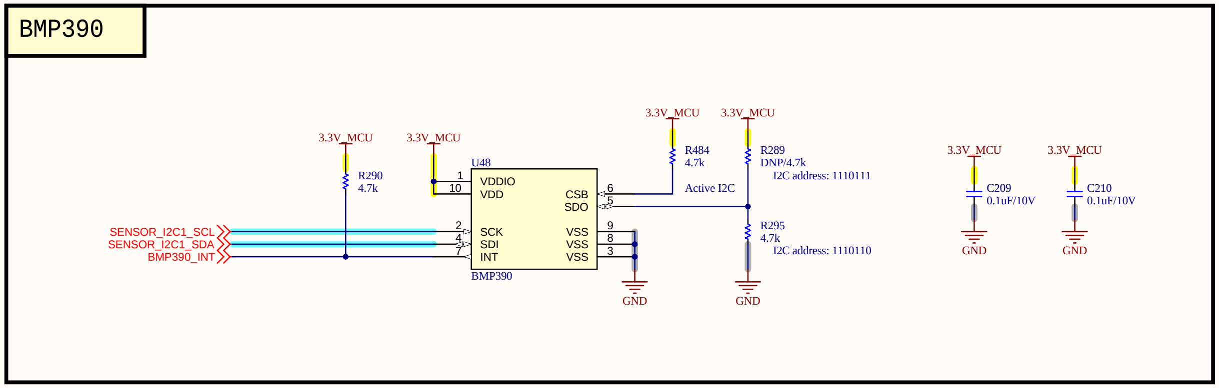

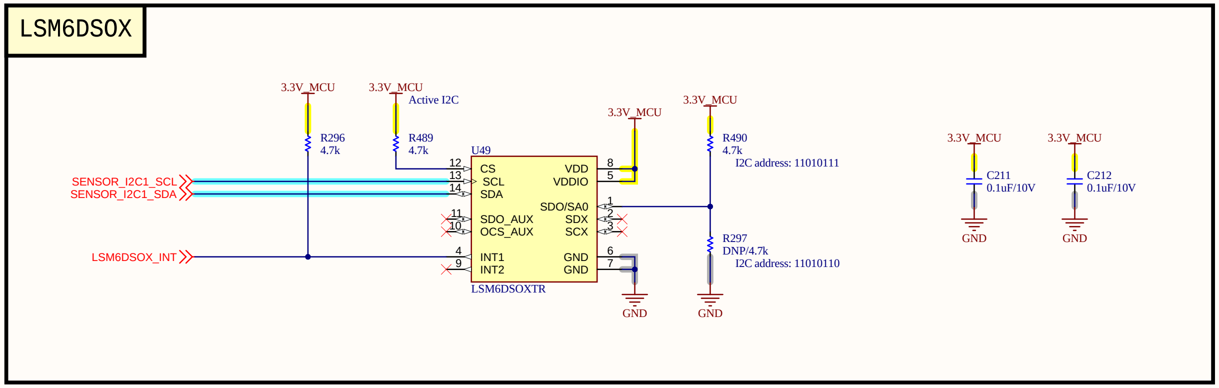

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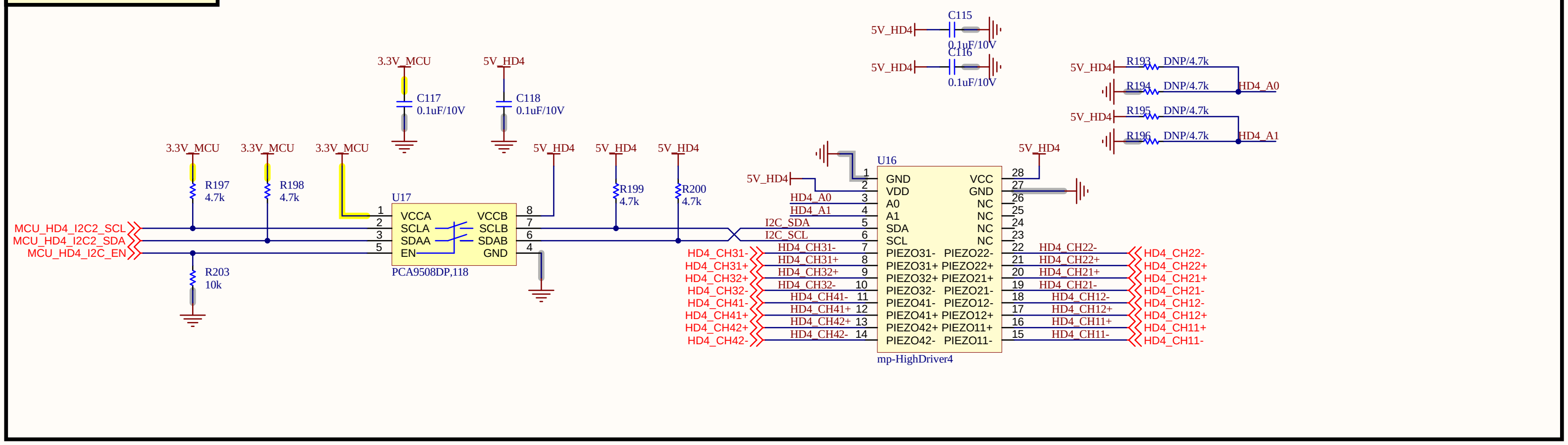
HIGH DRIVER 4



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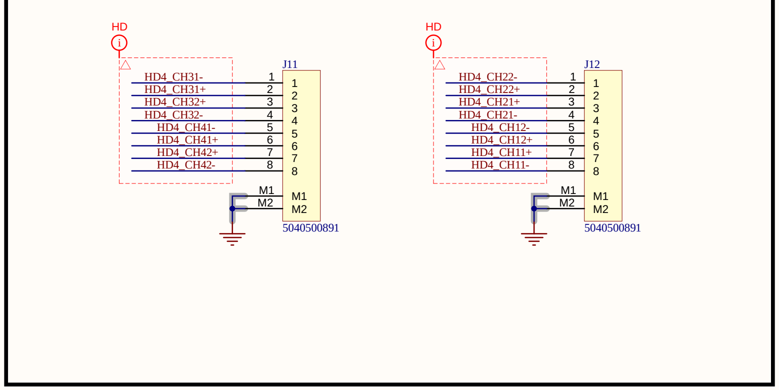

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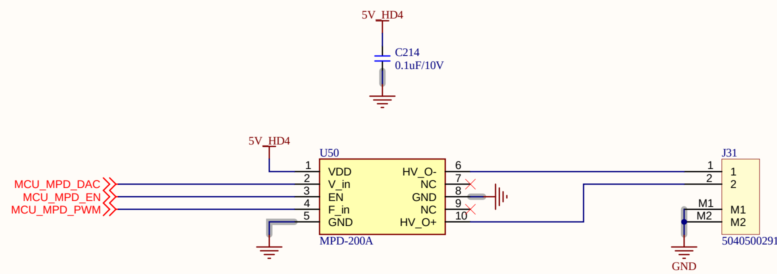

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Design by: SANG HUYNH	Sheet title: 18_HD4_PUMP_2.SchDoc	Sheet 29 of 33	
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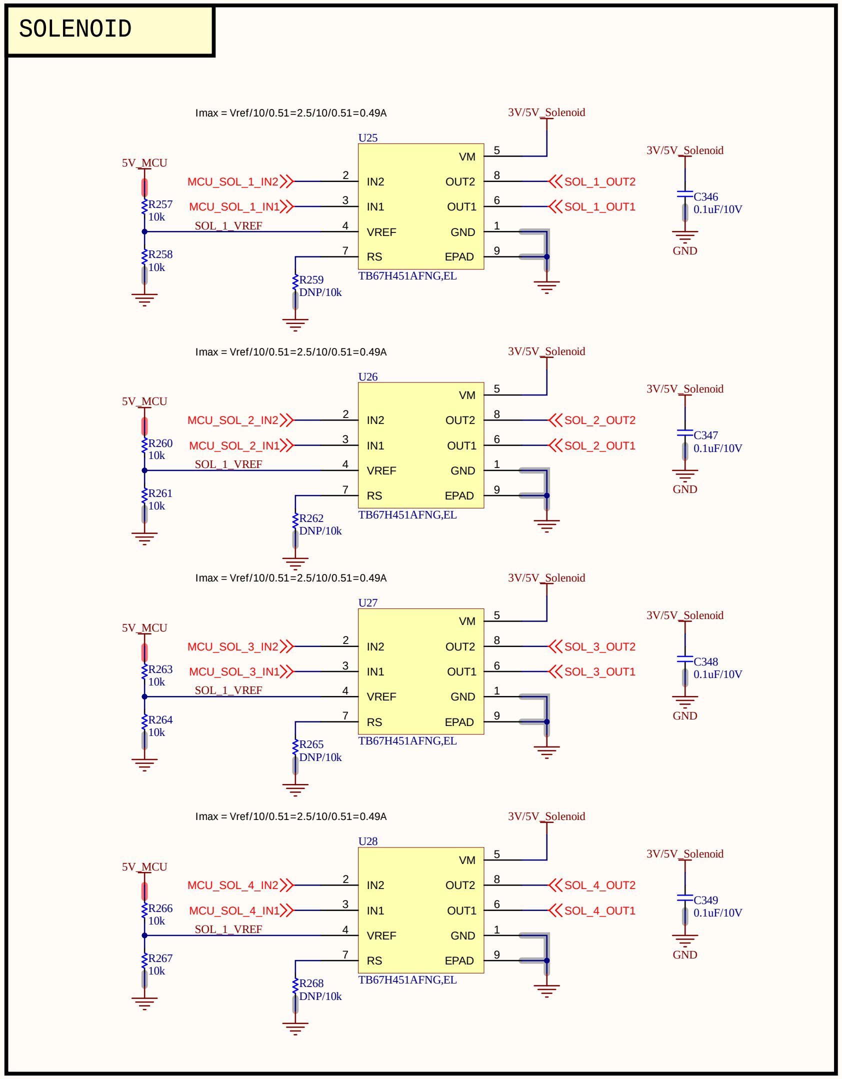

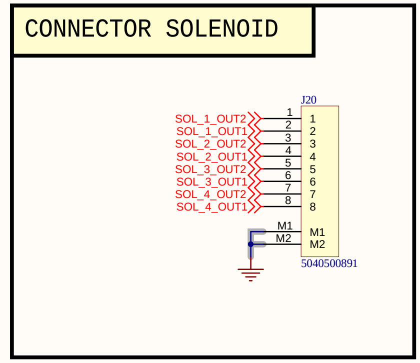

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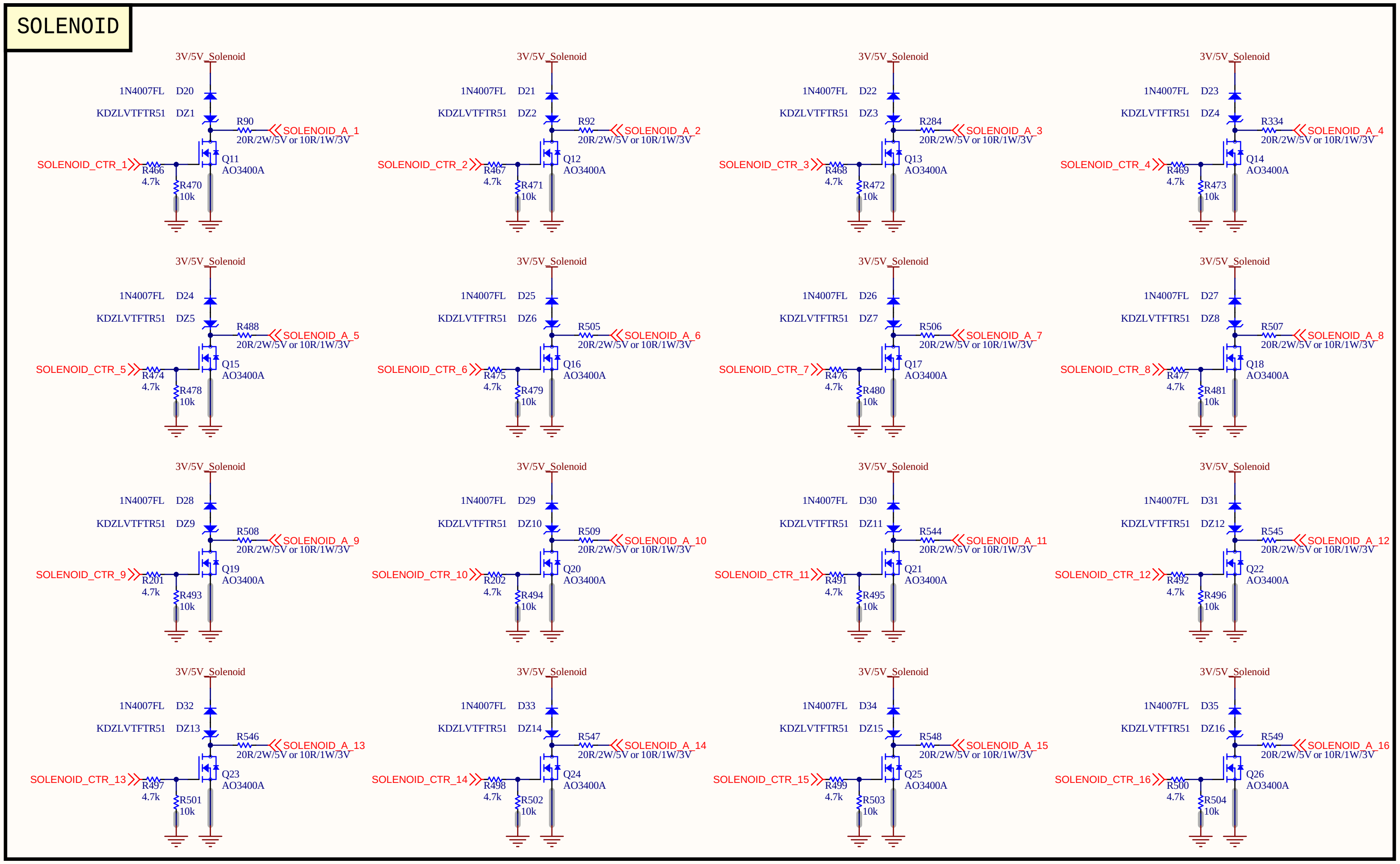
SOLENOID



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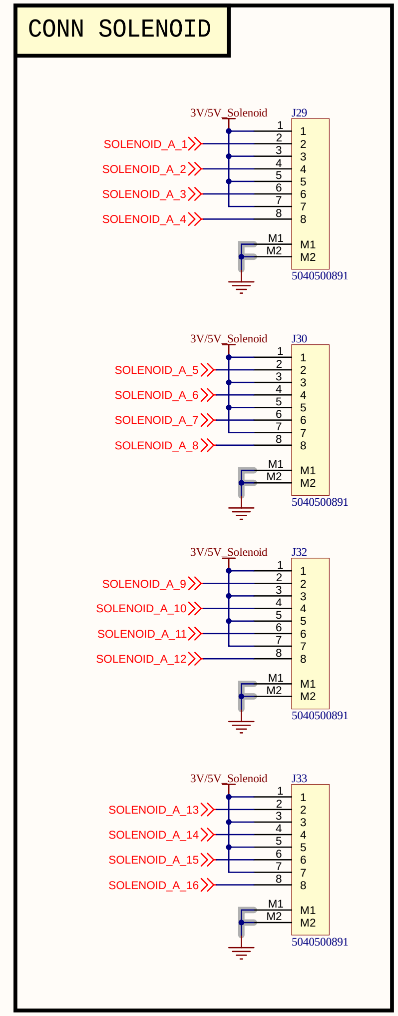

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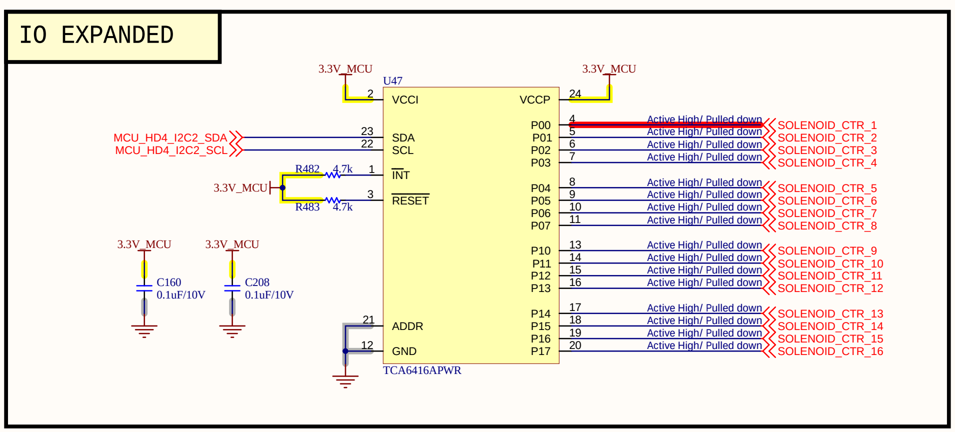

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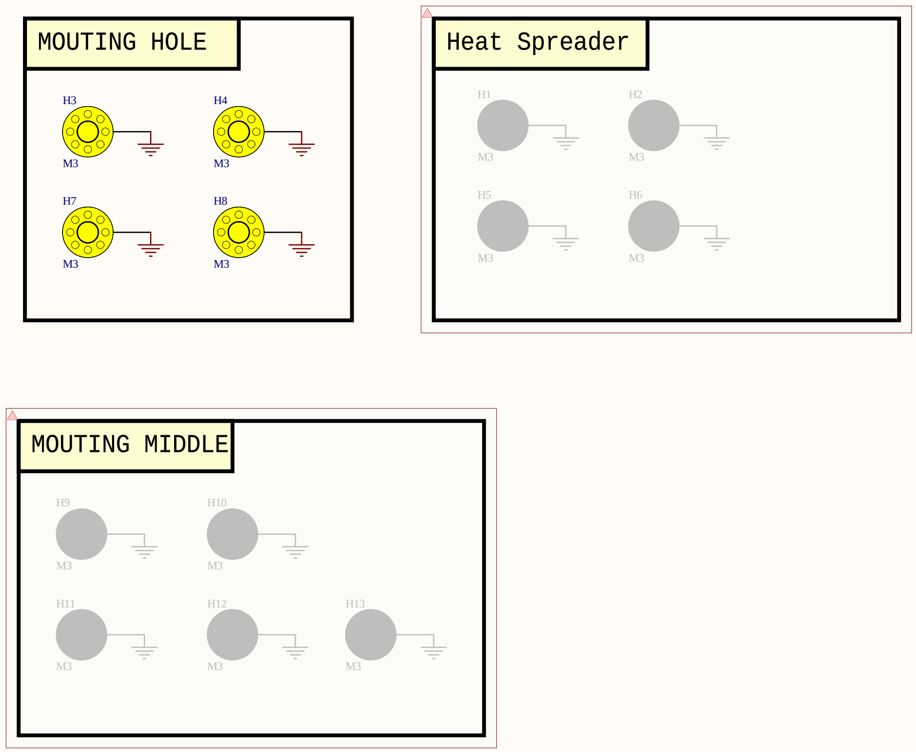

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Design by: SANG HUYNH	Sheet title: MOUTING.SchDoc	Sheet 32 of 33
Checked by: BAO BUI Q.	Document No: 32	Size: A3
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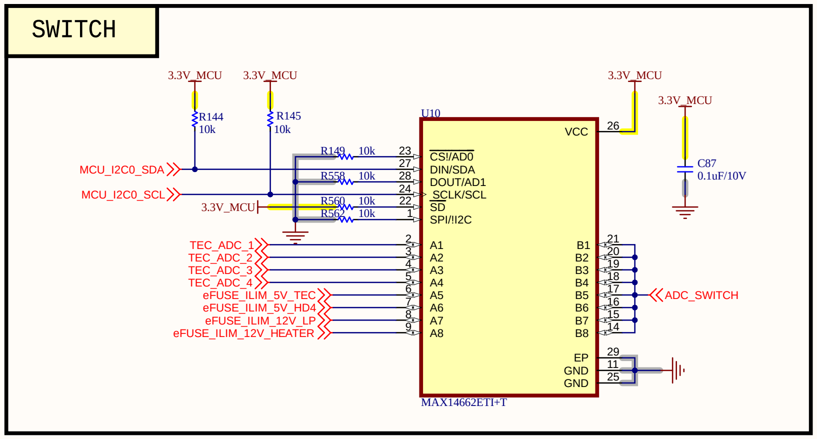

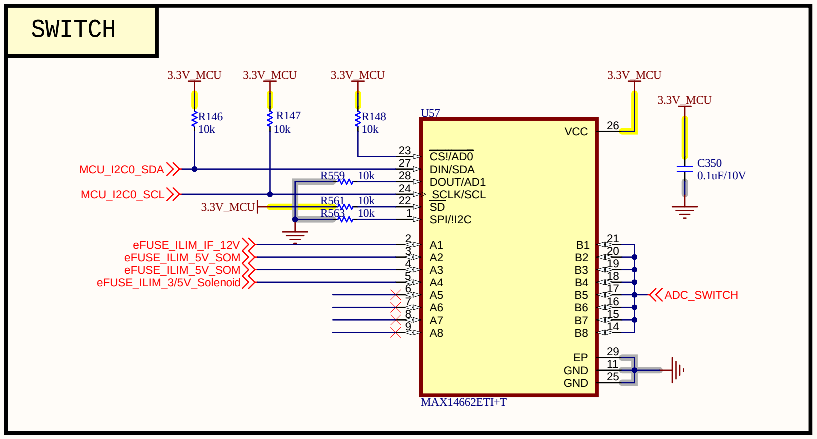

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